The University of Melbourne

An Integrated Framework for Blood Rheology Analysis

Final Project Report

UoM Group 35

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Abstract

Reliable detection of changes in blood rheological behaviour is important for monitoring blood properties and supporting the development of rheological measurement systems. This project, conducted in collaboration with Haemograph, investigated conventional rheological analysis, physics-based modelling, and signal-processing techniques for identifying changes in blood rheology.

Experimental measurements from the APR18 dataset were analysed alongside controlled synthetic and semi-synthetic signals with known change points. Conventional analysis examined rheological indicators, while the tensorial enhanced structural stress thixotropic-viscoelastic (t-ESSTV) model was implemented to predict stress responses. Fourier-based methods (FFT and STFT) and wavelet-based methods (CWT and DWT) were evaluated for their ability to characterise and detect signal changes.

The t-ESSTV model closely reproduced steady-state measurements but showed limitations in transient predictions. STFT and CWT detected controlled changes with feature-dependent differences in detection timing, while DWT provided useful multiscale representations without establishing consistent superiority over simpler measures. CWT showed promising results for localised change detection, although no single method demonstrated overall superiority.

The findings highlight the complementary roles of physical interpretation and signal analysis. CWT is recommended for further validation, while independent ground-truth measurements and broader noise-robustness testing remain necessary before practical deployment.

Declaration

We declare that this report is the original work of Project Group 35 and has been prepared collaboratively by its members as part of the Capstone Industry Project at the University of Melbourne.

All external sources, including published research, methodologies, and other referenced materials, have been appropriately acknowledged and cited.

Any use of generative artificial intelligence tools in preparing this report has been acknowledged in accordance with the University's academic integrity requirements. Details of their use are provided in the Generative AI Usage Declaration.

Project Group 35

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Contents

Abstract i

Declaration ii

1 Introduction 1
- 1.1 Industry Context and Project Motivation . . . 1
- 1.2 Background on Blood Rheology and Coagulation . . . 1
- 1.3 Problem Statement . . . 1
- 1.4 Project Objectives and Research Questions . . . 2
- 1.5 Scope of the Project . . . 2

2 Literature Review 2
- 2.1 Blood Rheological Behaviour . . . 2
- 2.2 Physical Modelling of Blood . . . 3
- 2.3 Interpretation of Nonlinear and Time-Dependent Responses . . . 4

3 Data and Methods 4
- 3.1 Data Overview . . . 4
- 3.2 Data Preparation and Preprocessing . . . 5
- 3.3 Overall Analysis Framework . . . 5
- 3.4 Evaluation Protocol . . . 5

4 Conventional Blood Rheology Analysis 6
- 4.1 Conventional Rheological Measurements . . . 6
- 4.2 Storage Modulus and Loss Modulus . . . 6
- 4.3 Gel-Point Detection and the Winter–Chambon Criterion . . . 7
- 4.4 Conventional Analysis Results . . . 8
- 4.5 Strengths and Limitations of Conventional Analysis . . . 10

5 Physical Modelling 10
- 5.1 t-ESSTV Model Implementation and Calibration . . . 10
- 5.2 APR18 Model Calibration and Evaluation . . . 12
- 5.3 APR18-Based Semi-Synthetic LAOS Evaluation . . . 14
- 5.4 Interpretation and Limitations . . . 17

6 Fast Fourier Transform and Short-Time Fourier Transform 17
- 6.1 FFT and STFT Analysis Methods . . . 17
- 6.2 Spectral Structure and Synthetic Feature Responses . . . 19
- 6.3 Synthetic Detection and Parameter Sensitivity . . . 21
- 6.4 Original APR18 Feature Variation and Reference Departures . . . 22
- 6.5 Controlled Perturbations and Noise Sensitivity . . . 23
- 6.6 Summary and Limitations . . . 24

7 Continuous Wavelet Transform (CWT) 24
7.1 CWT Method and Analysis Settings . . . 25
7.2 Time–Frequency Feature Extraction . . . 25
7.3 Controlled Gain-Change Experiments . . . 26
7.4 Change-Detection Performance and Feature-Specific Trade-offs . . . 26
7.5 Comparison with STFT . . . 28
7.6 CWT Analysis of the Original APR18 Data . . . 29
7.7 CWT Results Summary . . . 30

8 Discrete Wavelet Transform (DWT) 31
8.1 Aim, Data and Method . . . 31
8.2 Synthetic Results: Growth, Noise and Onset . . . 32
8.3 Real APR18 STEP: Trend and Sampling Limits . . . 33
8.4 Real APR18 LAOS: Periodic Components and Fast Activity . . . 34
8.5 Level Selection, Validation and Main Conclusion . . . 35

9 Conclusion 36
9.1 Main Findings . . . 36
9.2 Comparison and Recommended Approach for Haemograph . . . 36
9.3 Limitations . . . 37
9.4 Future Work . . . 37

10 Data Science Contributions 37
10.1 Technical Contributions . . . 37
10.2 Reproducibility and Project Handover . . . 37

References 39

A Generative AI Usage Declaration 41

1 Introduction

1.1 Industry Context and Project Motivation

This project was undertaken in collaboration with Haemograph Pty Ltd to investigate computational approaches for analysing blood rheology. The practical motivation is to extract interpretable information from measured stress responses and understand how those responses evolve under different loading conditions. Both physical modelling and signal analysis offer useful perspectives: constitutive models describe material behaviour through assumed physical mechanisms, while signal features summarise patterns observed in the measurements.

The project brings these perspectives together through a reproducible analysis workflow. Particular attention is given to time-dependent behaviour, the interpretation of nonlinear responses, and the practical limitations of identifying changes from finite, noisy records.

1.2 Background on Blood Rheology and Coagulation

Blood is a complex suspension whose mechanical response depends on its constituents and their interactions. Red blood cells can form reversible aggregates known as rouleaux, and their formation and breakdown contribute to time-dependent rheological behaviour. Blood can exhibit shear thinning, viscoelasticity and thixotropy, motivating models that account for both deformation and structural evolution[3, 5].

These properties make stress responses dependent on loading conditions and deformation history. Steady-state, step-shear and oscillatory experiments therefore provide different information about material behaviour. Thixo-elasto-visco-plastic models seek to describe these interacting effects within a constitutive framework[4, 14].

Although coagulation motivates interest in changing blood properties, reversible aggregation, a measured stress departure and clot formation are not interchangeable observations. Identifying a change in a mechanical signal does not, by itself, establish its physiological cause.

1.3 Problem Statement

The central problem is how to describe and identify meaningful changes in blood stress signals while separating them from imposed loading, ordinary background variation and measurement-related effects. An overall increase in stress may reflect amplitude scaling, whereas changes in relative harmonics or time-scale components may indicate different signal behaviour. These interpretations require appropriate comparisons rather than relying on visual differences alone.

Temporal analysis introduces further constraints. Longer windows can improve spectral estimation but combine observations over a wider interval. Shorter windows may respond sooner while increasing estimation error. Reference selection, thresholds and persistence rules also affect whether and when a departure is confirmed.

The available experimental records do not provide verified material-change times. Real-data analysis can therefore support description and consistency checks, but labelled synthetic experiments are needed for controlled evaluation.

1.4 Project Objectives and Research Questions

The project aims to assess complementary methods for describing blood rheology and investigating signal changes. Its objectives are to:

1. Examine conventional rheological measures and implement selected physical models to assess their agreement with measured stress responses.
2. Extract interpretable frequency and time-scale features using FFT, STFT, CWT and DWT.
3. Evaluate feature responses and, where a detection rule is defined, confirmation timing under controlled synthetic changes and noise.
4. Assess sensitivity to analysis settings and provide reproducible code and outputs for further investigation by the client.

The corresponding research questions concern what each method reveals, whether it adds information beyond simpler signal summaries, and which limitations affect its use for change analysis. Comparisons distinguish descriptive interpretability, numerical estimation and event timing, rather than assuming a single method is best for every task.

1.5 Scope of the Project

The study covers supplied experimental blood-rheology records and controlled synthetic experiments. Physical modelling examines stress responses under selected protocols, while Fourier and wavelet analyses investigate harmonic content, local variation and multiscale structure.

The methods do not all use identical datasets or evaluation rules. Direct performance comparisons are therefore limited to matched experiments, with other findings interpreted within their respective settings. Known synthetic onsets refer to imposed mechanical changes rather than verified coagulation events.

The project provides an exploratory research workflow, not a clinically validated diagnostic system. Its deliverables include implementations, quantitative comparisons, documented settings and reusable analysis outputs. Physiological interpretation and generalisation to independent blood samples require additional validation.

2 Literature Review

2.1 Blood Rheological Behaviour

Blood exhibits several interacting rheological properties, including shear thinning, viscoelasticity and thixotropy. These behaviours are related to its cellular composition and the formation and breakdown of red blood cell aggregates, known as rouleaux. Understanding this relationship between microstructure and stress response is central to the constitutive models considered in this review[3, 5].

Shear thinning describes a decrease in apparent viscosity with increasing shear rate. Thixotropy concerns the time-dependent evolution of structure under an imposed flow history. These effects

are related but distinct: a steady flow curve describes shear-rate dependence, whereas transient measurements reveal how the response evolves towards a new state. Structural evolution models represent this behaviour through competing mechanisms of rouleaux breakdown and rebuilding[3].

Viscoelasticity introduces both recoverable deformation and viscous dissipation. The selected studies also incorporate yielding and plastic deformation into their descriptions of blood. The resulting thixo-elasto-visco-plastic frameworks seek to represent these effects together, rather than treating viscosity as a function of the instantaneous shear rate alone[4, 14].

2.2 Physical Modelling of Blood

The selected studies share the aim of connecting time-dependent blood behaviour with evolving structure, but differ in how they formulate stress and combine its physical contributions.

Armstrong and Pincot[3] extended the Giesekus model by adding elastic and viscoelastic stress contributions associated with rouleaux. These contributions were superimposed on a viscoelastic backbone, while a structural parameter described aggregate formation and breakdown. The framework was evaluated against oscillatory measurements and compared with other enhanced thixotropic models. Its performance varied with the flow condition, illustrating that a more complex constitutive equation does not automatically provide a better description of every experiment.

Armstrong et al.[4] followed another approach by incorporating thixotropy into an elasto-visco-plastic model. A structural evolution term allowed the model to represent changes in microstructure alongside elastic, viscous and plastic behaviour. Step-up and step-down shear experiments were used for parameter fitting, with oscillatory experiments providing further model comparisons. This work highlights the importance of testing time-dependent models beyond steady-state flow curves.

Pincot and Armstrong[14] developed the tensorial-enhanced-thixo-visco-plastic (t-e-TVP) model by combining elements of Bingham and generalised Maxwell systems with structural stress contributions. The tensorial formulation represents stress components beyond a single scalar shear stress. The model was compared with the tensorial enhanced thixotropic mHAWB framework using small-amplitude, large-amplitude and unidirectional oscillatory shear. The comparison considered predictive capability alongside the number of model parameters, highlighting the balance between model complexity and descriptive performance.

Armstrong et al.[5] examined the tensorial enhanced structural stress thixotropic-viscoelastic (t-ESSTV) model. This framework distinguishes viscoelastic responses associated with individual blood constituents and rouleaux from the timescales of structural breakdown and aggregation. Separating these mechanisms provides a physical basis for interpreting transient responses that cannot be described by a single relaxation process.

Together, these studies show that model selection involves more than matching an overall stress curve. The mechanisms represented, parameter requirements and range of experimental validation also determine how a model can be interpreted and applied.

2.3 Interpretation of Nonlinear and Time-Dependent Responses

Oscillatory and transient experiments reveal different aspects of blood rheology. Small-amplitude oscillatory shear characterises approximately linear viscoelastic behaviour, whereas large-amplitude oscillatory shear (LAOS) probes nonlinear responses. Under LAOS, higher harmonics and changes within an oscillation cycle contain information that is not fully expressed by conventional linear storage and loss moduli[5].

Armstrong et al.[5] combined t-ESSTV modelling with the Sequence of Physical Processes (SPP) framework. SPP evaluates the changing elastic and viscous response throughout an oscillation, rather than assigning only a cycle-level description. The study compared these measures with model-predicted rouleaux structure, connecting observable mechanical responses with a proposed microstructural interpretation.

Hysteresis provides another view of time-dependent behaviour. During an increasing and decreasing shear-rate ramp, stress can follow different paths at the same shear rate because the structural state depends on the preceding deformation history. Pincot et al.[15] investigated this behaviour using triangular shear-rate ramps and related it to the different timescales of rouleaux breakdown and aggregation.

Their study fitted t-ESSTV parameters using steady-state and hysteresis measurements, then used those parameters to predict step-shear and oscillatory responses. This separates agreement with calibration measurements from performance under other flow histories. Hysteresis-loop analysis therefore complements oscillatory tests by examining how structural evolution affects the response during changing shear conditions[15].

Overall, the five studies provide complementary perspectives on blood rheology: constitutive models describe interacting stress contributions, while oscillatory and hysteresis analyses examine how those contributions evolve under deformation. Model-derived structural variables remain interpretations within the chosen framework, rather than direct observations of microstructure. Their usefulness should therefore be assessed together with agreement across relevant experimental conditions.

3 Data and Methods

3.1 Data Overview

The study primarily used the APR18 experimental blood rheology dataset, corresponding to Donor 5 in Armstrong et al. [5]. The dataset contains steady-state (SS), step-shear, small-amplitude oscillatory shear (SAOS), and large-amplitude oscillatory shear (LAOS) measurements. These protocols capture rheological behaviour under constant, changing, and oscillatory shear conditions.

To address the absence of known change points in the experimental data, controlled semi-synthetic signals were constructed from APR18 LAOS measurements. We labelled $\gamma = 10$ as a common background. It includes an unchanged control and three cases with final stress gains

of 10%, 20%, and 30%. The injected signal is

$$\sigma_\alpha(t) = \sigma_{\mathrm{APR18}}(t)[1 + \alpha q(t)], \qquad q(t) = \begin{cases} 0, & t < 15, \\ 1 - \exp[-((t-15)/5)^2], & t \geq 15, \end{cases} \tag{1}$$

with t in seconds and $\alpha \in \{0.10, 0.20, 0.30\}$. Measured strain and background variability are retained, and no extra noise is added. The ground truth is the onset of this prescribed mechanical perturbation at 15 s.

Figure 13 shows the feature responses. In the late window $[26.00, 38.56)$ s, the harmonic amplitudes increase by approximately the imposed gains. The power proxy increases by 20.99%, 43.98%, and 68.97%, close to the constant-gain prediction $(1+\alpha)^2 - 1$. Both harmonic ratios change by less than 0.008% relative to their matched-control values in this window.

3.2 Data Preparation and Preprocessing

Experimental records were organised by measurement protocol. Relevant time, stress, strain, and shear-rate variables were extracted and checked for missing or invalid values. Time series were aligned using their recorded timestamps, with preprocessing adapted to the requirements of each analytical method.

For physical modelling, measured shear-rate histories were interpolated to provide continuous inputs for numerical integration. Signal-processing analyses used the recorded time-series measurements to examine changes in their temporal and frequency-domain characteristics.

3.3 Overall Analysis Framework

Three complementary approaches were investigated. Conventional rheological analysis examined changes in established mechanical indicators, including storage and loss moduli. Physical modelling employed t-ESSTV to predict stress responses and assess deviations between model predictions and observations. Signal-processing methods examined changes in frequency content and localised signal characteristics.

The results from these approaches were evaluated to determine their respective capabilities and limitations for identifying rheological changes.

3.4 Evaluation Protocol

The three approaches were evaluated based on their ability to identify changes in rheological behaviour. For controlled synthetic or semi-synthetic data with known change points, detection performance was assessed using detected onset, detection delay, and false alarms where applicable. For experimental data without verified change points, the analysis focused on changes in measured indicators and agreement between methods rather than confirmed detection accuracy. Method-specific evaluation metrics and detection criteria are described in the corresponding sections.

4 Conventional Blood Rheology Analysis

Conventional rheological analysis focuses on the observation of blood behaviour before, during and after cogulation. As blood is a viscoelastic material, meaning that it behaves as both a viscous liquid and elastic solid, being able to identify interesting features of each stage of the blood flow process can be useful in many fields. [8, 17].

This section introduces conventional rheological measurements using a rheometer, the interpretation of storage and loss moduli, and established approaches for identifying the gel point. The methods are then applied to synthetic experimental rheological data as a baseline to evaluate their effectiveness in detecting blood coagulation.

4.1 Conventional Rheological Measurements

Rheometry is commonly used to characterise the mechanical properties of blood by applying a controlled deformation and measuring the corresponding stress response [8].

Two commonly used oscillatory testing approaches for blood are small-amplitude oscillatory shear (SAOS) and large-amplitude oscillatory shear (LAOS).

Under oscillatory shear, the applied strain can be expressed as

$$\gamma(t) = \gamma_0 \sin(\omega t), \tag{2}$$

where γ_0 denotes the strain amplitude, ω is the angular frequency, and t represents time.

For a linear viscoelastic response, the resulting stress is

$$\sigma(t) = \sigma_0 \sin(\omega t + \delta), \tag{3}$$

where σ_0 represents the stress amplitude and δ is the phase difference between stress and strain.

In SAOS measurements, sufficiently small deformations are applied to characterise the linear viscoelastic response, allowing the measurements to be observed while minimizing any mechanical disturbances. In contrast, LAOS measurements involve larger deformations that may produce nonlinear stress responses containing higher harmonics, but may introduce more noise to the signal in the process. These techniques provide information about changes in the mechanical behaviour of blood throughout the coagulation process.

4.2 Storage Modulus and Loss Modulus

The viscoelastic response of blood is commonly characterised using the storage modulus (G') and loss modulus (G''), which represent the elastic and viscous contributions, respectively.

For linear oscillatory shear, the complex shear modulus is

$$G^*(\omega) = G'(\omega) + iG''(\omega), \tag{4}$$

where i denotes the imaginary unit.

The storage modulus is calculated as

$$G' = \frac{\sigma_0}{\gamma_0}\cos(\delta), \tag{5}$$

while the loss modulus is

$$G'' = \frac{\sigma_0}{\gamma_0}\sin(\delta). \tag{6}$$

The storage modulus quantifies the energy stored and subsequently recovered during each oscillation, whereas the loss modulus characterises the energy dissipated through viscous processes.

The relative contribution of these two components can be expressed using the loss tangent,

$$\tan(\delta) = \frac{G''}{G'}, \tag{7}$$

which indicates whether the material exhibits predominantly viscous or elastic behaviour.

When $G'' > G'$, the material demonstrates a predominantly viscous response. Conversely, when $G' > G''$, elastic behaviour becomes dominant.

During coagulation, the formation and development of a fibrin network can alter these viscoelastic properties, providing measurable indicators of clot development [17, 6].

4.3 Gel-Point Detection and the Winter–Chambon Criterion

The gel point represents the transition at which a system-spanning network forms, marking the development of a mechanically connected clot structure.

In conventional rheological analysis, a commonly examined indicator is the crossover between the storage and loss moduli,

$$G'(t_c) = G''(t_c), \tag{8}$$

where t_c denotes the crossover time.

Although this crossover can indicate a transition in the relative dominance of viscous and elastic behaviour, it does not necessarily correspond to the true gel point.

A more rigorous approach is provided by the Winter–Chambon criterion, which identifies the critical gel state through the frequency dependence of the viscoelastic moduli.

At the gel point, the moduli exhibit power-law behaviour,

$$G'(\omega) \propto \omega^n, \qquad G''(\omega) \propto \omega^n, \tag{9}$$

where n is the critical relaxation exponent, with $0 < n < 1$.

Consequently, the loss tangent becomes independent of frequency,

$$\tan(\delta) = \frac{G''(\omega)}{G'(\omega)} = \tan\left(\frac{n\pi}{2}\right). \tag{10}$$

The gel point can therefore be identified as the time at which the loss tangent becomes approximately frequency-independent across multiple measurement frequencies.

This principle has been applied to the rheological characterisation of blood clot formation [6, 7].

4.4 Conventional Analysis Results

Conventional rheological analysis was performed to investigate changes in the mechanical properties of blood during coagulation.

The analysis focused on the evolution of storage and loss moduli, changes in the loss tangent, and potential indicators of gel formation.

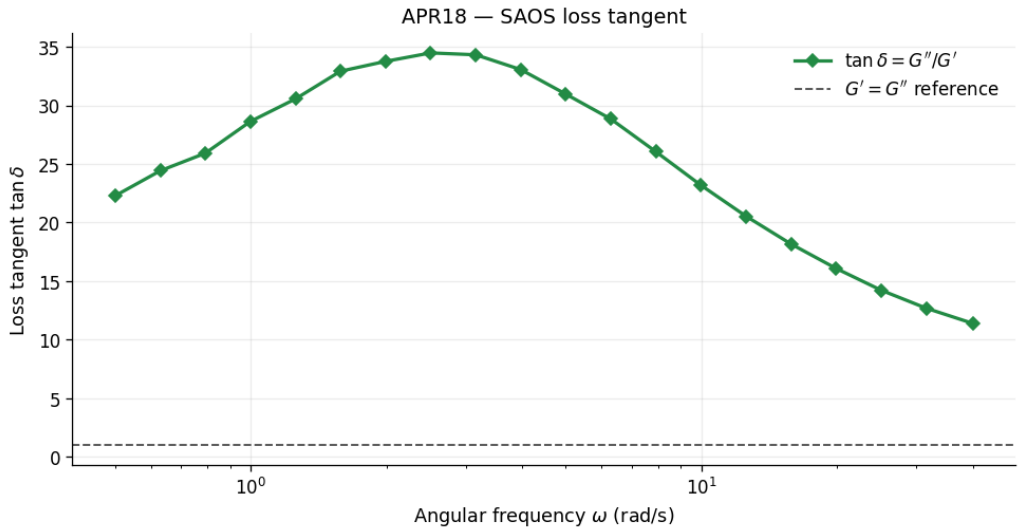


Figure 1: Loss tangent of the APR18 SAOS frequency sweep, illustrating the frequency-dependent viscoelastic behaviour of the blood sample.

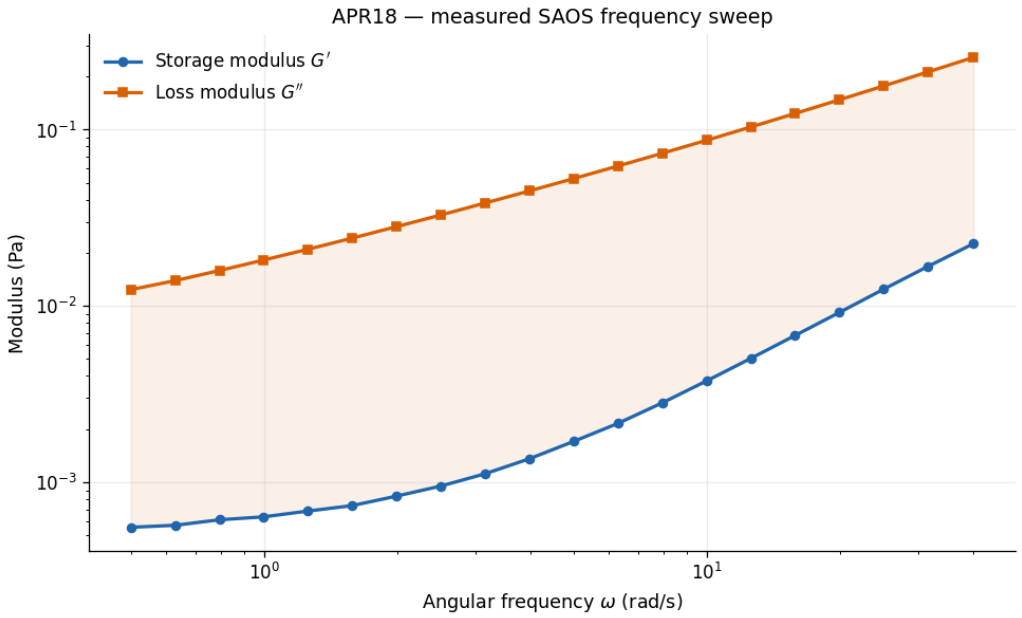


Figure 2: Storage modulus G' and loss modulus G'' of APR18 measured using SAOS across different angular frequencies.

4.4.1 SAOS Analysis

SAOS analysis was conducted to characterise the linear viscoelastic behaviour of blood using the APR18 experimental dataset. As illustrated in Figure 2, the loss modulus (G'') remains consistently greater than the storage modulus (G') across the measured frequency range, indicating predominantly viscous behaviour. Although both moduli increase with angular frequency, no crossover is observed, suggesting that the measured sample does not exhibit a transition to elastic-dominated behaviour under these experimental conditions.

Furthermore, Figure 1 illustrates the frequency dependence of the loss tangent ($\tan\delta = G''/G'$). The loss tangent remains greater than unity throughout the frequency sweep, further confirming viscous-dominated behaviour. Its variation across frequencies also indicates that the measurements do not satisfy the frequency-independent loss tangent expected at the critical gel point under the Winter–Chambon criterion.

While SAOS provides valuable information regarding the linear viscoelastic properties of blood, it is limited to measurements within the linear viscoelastic regime and therefore cannot fully characterise nonlinear responses under larger deformations. Consequently, LAOS analysis was also conducted to investigate the nonlinear viscoelastic behaviour of blood and evaluate potential indicators of clot formation.

4.4.2 LAOS Analysis

Complementing SAOS analysis, LAOS provides insight into the nonlinear viscoelastic behaviour of blood. However, the larger imposed deformations associated with LAOS may alter the structure of the blood sample and influence the measured response.

To identify potential gelation transitions using LAOS data, we focus on the crossover between the first-harmonic storage modulus G'_1 and loss modulus G''_1, rather than analysing the entire oscillatory signal directly.

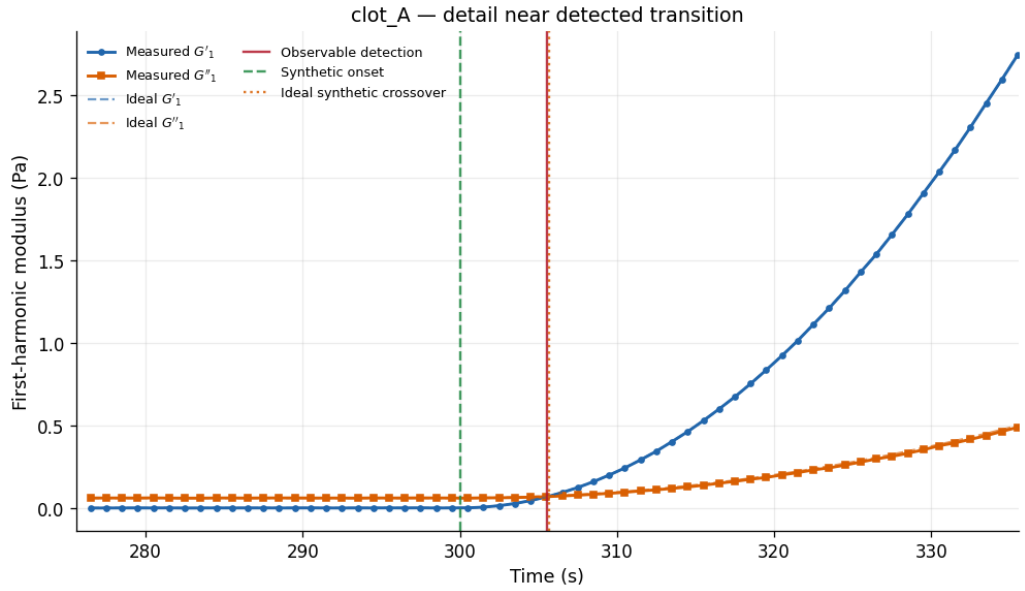


Figure 3: Conventional LAOS FFT analysis of synthetic clotting sample clot_A, showing the first-harmonic storage and loss moduli and their crossover during the injected clotting transition.

Table 1: Comparison of injected clotting onset, ground-truth crossover time, and detected crossover time using conventional LAOS FFT analysis.

Sample	Onset (s)	Ground Truth (s)	Detected Time (s)
clot_A	300.04	305.67	305.57
clot_B	240.04	246.09	246.01
clot_C	360.04	365.33	365.29

4.5 Strengths and Limitations of Conventional Analysis

It is clearly shown that when a clot is present, in this case injected synthetically, the conventional analysis is able to provide us with a slight delayed detection of the clot. This leads to the conclusion that we have a strong criterion for detecting a blood transitional point.

Conventional rheological analysis provides several advantages for studying blood coagulation.

First, measurements such as the storage modulus, loss modulus, and loss tangent have clear physical interpretations, allowing changes in blood structure to be related to measurable mechanical properties.

Second, established criteria for gel-point identification provide a theoretical basis for characterising the transition from a predominantly fluid-like state to a mechanically connected network [6].

However, conventional approaches also present limitations.

A modulus crossover does not necessarily identify the true gel point, particularly when measurements are performed at only a single oscillation frequency.

Furthermore, conventional analysis based on measurements averaged over oscillation cycles may provide limited temporal resolution when material properties change rapidly during coagulation.

The Winter–Chambon criterion also requires measurements over multiple frequencies, which may introduce experimental challenges when analysing a rapidly evolving material.

Consequently, although conventional rheological methods provide valuable physical information, additional signal-processing approaches may be useful for investigating the temporal evolution of coagulation.

This motivates the application of time-frequency analysis techniques, which aim to characterise time-dependent changes in rheological signals [20].

5 Physical Modelling

5.1 t-ESSTV Model Implementation and Calibration

The tensorial enhanced structural stress thixotropic-viscoelastic (t-ESSTV) model was implemented to investigate whether physics-based shear-stress predictions could support change detection in blood rheology. Following Armstrong et al. [5], the model combines the viscoelastic contribution of plasma and deformable red blood cells with the evolving stress contribution of rouleaux microstructure. A structural parameter, λ, represents the degree of rouleaux aggre-

gation, ranging from zero (unaggregated) to one (fully structured). The total shear stress is expressed as

$$\sigma_{\mathrm{tot},yx} = \sigma_{yx,\mathrm{ve}} + \sigma_{yx,\mathrm{R,evp}}, \tag{11}$$

where the two terms represent the viscoelastic stress contribution of plasma and individual red blood cells, and the elastoviscoplastic contribution of rouleaux, respectively. The model predicts stress by integrating coupled stress and structural-evolution equations under an imposed shear-rate history. The full tensorial equations and parameter definitions are provided by Armstrong et al. [5].

Model parameters were calibrated using the APR18 experimental dataset. The calibration data consisted of 16 steady-state measurements and six step-shear protocols: $0.1 \rightarrow 1$, $0.1 \rightarrow 2.5$, $0.1 \rightarrow 5$, $5 \rightarrow 0.5$, $5 \rightarrow 1$, and $5 \rightarrow 2.5$ s^{-1}. The $0.1 \rightarrow 0.5$ s^{-1} protocol was excluded from calibration for subsequent evaluation. Measured shear-rate histories were interpolated piecewise-linearly and supplied to the transient model rather than approximating the changes as instantaneous steps.

Following the staged approach described by Armstrong et al. [5], parameter estimation proceeded through scalar steady-state calibration, scalar transient calibration, and joint tensorial calibration. Our implementation used a replica-exchange (parallel-tempering) optimisation scheme to explore bounded parameter values, followed by local Powell refinement. The final tensorial stage optimised ten variable parameters. Two random seeds (11 and 29) were evaluated, with seed 29 selected based on its lower calibration objective (0.0111496 compared with 0.0111512). The held-out protocol was not used for parameter selection. Three additional parameters, $\gamma_{0,r} = 1$, $d = 0.5$, and $m = 1.5$, were fixed according to the settings reported by Armstrong et al. [5].

Table 2: Comparison of published Donor 5 t-ESSTV parameters from Armstrong et al. [5] and our fitted APR18 parameters.

Parameter	Published Donor 5	Our APR18 fit
$\mu_{0,c}$	0.0138	0.009674
$\mu_{\infty,c}$	0.0054	0.005288
τ_c	0.029	0.015317
τ_{break}	1.582	2.103705
τ_{aggr}	0.192	1.196066
μ_r	0.099	0.101964
σ_{y0}	0.0039	0.004749
τ_λ	1.4582	1.392394
G_r	0.2646	0.442179
G_c	0.5138	1.151511

Table 2 compares our fitted parameters with the published Donor 5 values. While several viscosity parameters were relatively similar, larger differences were observed in the aggregation timescale and elastic moduli. These discrepancies may reflect differences in calibration procedures, optimisation objectives, and parameter interactions. As our implementation does not exactly reproduce the published fitting procedure, the results are treated as an independent

evaluation of the t-ESSTV framework. The fitted parameters were subsequently held fixed for held-out protocol evaluation and LAOS prediction.

5.2 APR18 Model Calibration and Evaluation

5.2.1 Steady-State Calibration

The model was evaluated using 16 APR18 steady-state measurements. As shown in Figure 4, the fitted stress closely followed the experimental measurements across the investigated shear-rate range. The model achieved an RMSE of 0.00760 Pa and an R^2 of 0.99995, indicating excellent agreement with the calibration data. However, this reflects in-sample fitting performance rather than predictive accuracy on unseen measurements.

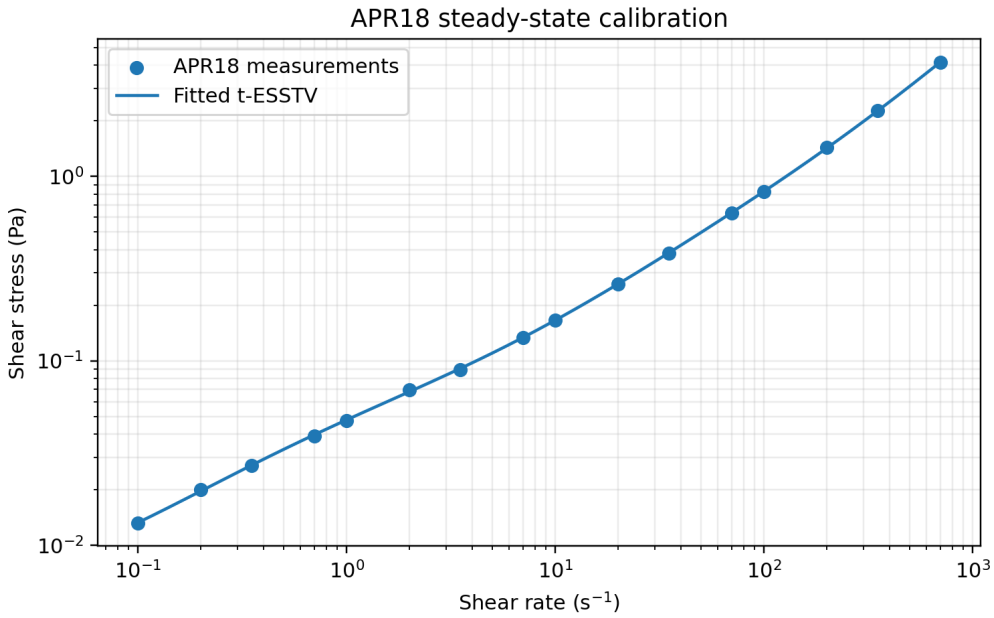


Figure 4: Steady-state calibration of the t-ESSTV model using 16 APR18 experimental measurements.

5.2.2 Transient Step-Shear Calibration

The model was further evaluated using six step-up and step-down shear protocols. Figure 5 compares measured and predicted stress responses. The model generally captured the direction and overall magnitude of the stress changes, with R^2 values ranging from 0.773 to 0.958. However, discrepancies remained in the transient responses. In several step-up protocols, the predicted stress relaxed more rapidly than the experimental measurements and slightly underestimated the final stress level. These findings suggest that the model captured the overall stress evolution but did not fully reproduce the transient relaxation dynamics.

5.2.3 Held-Out Protocol Prediction

To assess performance beyond the calibration data, the $0.1 \rightarrow 0.5$ s^{-1} step-up protocol was held out during parameter estimation. Using the fixed fitted parameters, the model achieved an

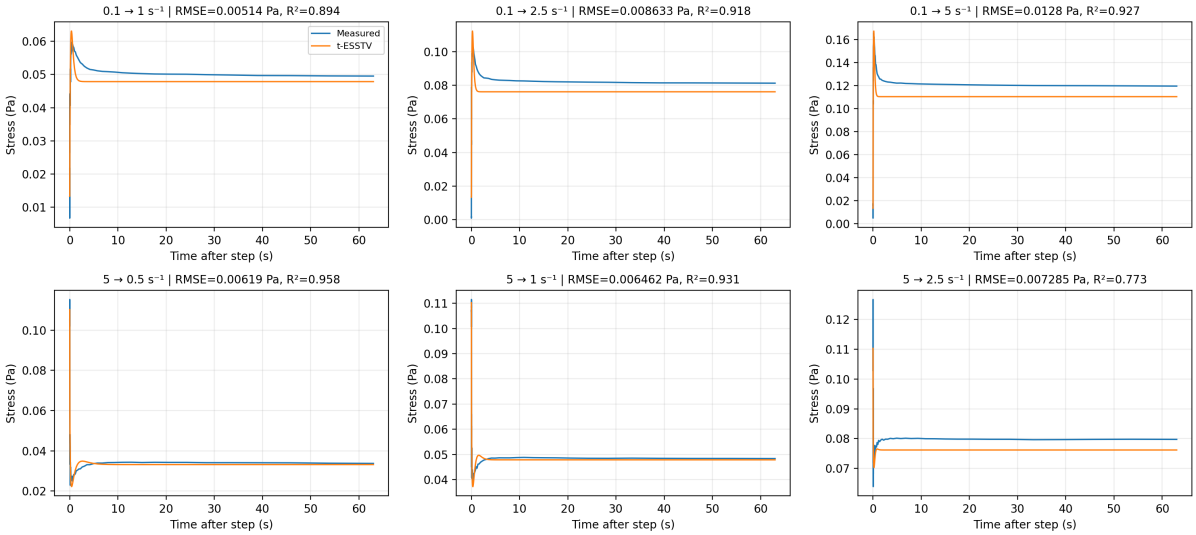


Figure 5: Measured and predicted stress responses for six APR18 step-shear calibration protocols.

RMSE of 0.00398 Pa and an R^2 of 0.839 (Figure 6). Although the prediction followed the overall stress trend, it exhibited faster relaxation and a lower final stress level than the experimental response.

This demonstrates predictive capability on one unseen protocol while highlighting limitations in modelling transient stress evolution.

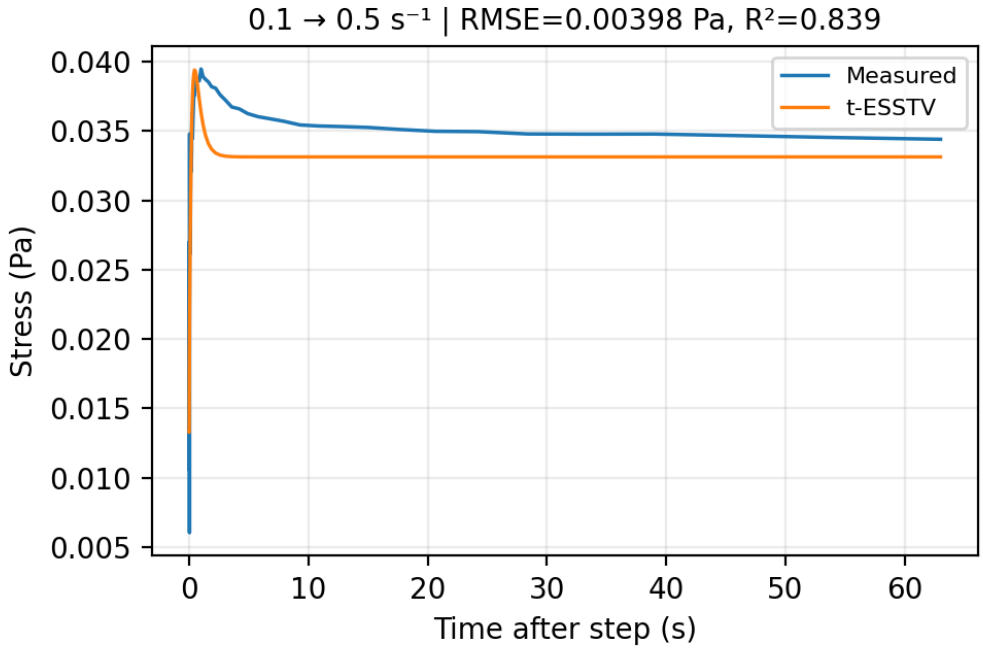


Figure 6: Held-out prediction for the APR18 $0.1 \rightarrow 0.5\ \mathrm{s}^{-1}$ step-up protocol using fixed t-ESSTV parameters.

5.3 APR18-Based Semi-Synthetic LAOS Evaluation

5.3.1 Controlled Perturbation Construction

To evaluate change detection with a known onset, a semi-synthetic benchmark was constructed from an experimentally recorded APR18 LAOS signal at angular frequency $\omega = 1\ \mathrm{rad\,s^{-1}}$ and strain amplitude $\gamma_0 = 10$. The record contained 1,995 observations over approximately 40 seconds, sampled at approximately 50 Hz. The unchanged experimental waveform served as the control, while three additional cases introduced gradual increases in stress amplitude of 10%, 20%, and 30%.

The perturbed signal was defined as

$$\sigma_\alpha(t) = \sigma_{\mathrm{APR18}}(t)\,[1 + \alpha q(t)]\,, \qquad \alpha \in \{0, 0.10, 0.20, 0.30\}, \tag{12}$$

where

$$q(t) = \begin{cases} 0, & t < 15\ \mathrm{s}, \\ 1 - \exp\left[-\left(\frac{t-15}{5}\right)^2\right], & t \geq 15\ \mathrm{s}. \end{cases} \tag{13}$$

The ground-truth onset was therefore 15 seconds for each perturbation case. The original time, strain, derived shear-rate history, and experimental variability were retained without adding random noise. These controlled perturbations were designed to evaluate sensitivity to changes in stress amplitude rather than to simulate confirmed coagulation events.

5.3.2 Frozen Stress Prediction and Detection Criterion

The calibrated tensorial model was applied to all four LAOS cases using the same fixed parameter set. Prediction errors were quantified using a rolling root mean squared error (RMSE):

$$R(t) = \sqrt{\frac{1}{|W_t|} \sum_{i \in W_t} \left[\sigma_{\mathrm{obs}}(t_i) - \hat{\sigma}_{\mathrm{tESSTV}}(t_i)\right]^2}, \tag{14}$$

where W_t represents a trailing two-second window of observations. This window smooths short-term fluctuations in the residual while preserving temporal information about changes. At $\omega = 1\ \mathrm{rad\,s^{-1}}$, one oscillation lasts approximately 2π seconds, so the two-second window is shorter than one complete cycle.

A common detection threshold was established from the control rolling RMSE:

$$T = \max_t R_{\mathrm{control}}(t) + 3 \times 1.4826 \times \mathrm{MAD}\left(R_{\mathrm{control}}\right), \tag{15}$$

where

$$\mathrm{MAD}(x) = \mathrm{median}\left(|x - \mathrm{median}(x)|\right). \tag{16}$$

The resulting threshold was $T = 0.0240178$ Pa and remained fixed across all perturbation cases. An alarm was triggered only when rolling RMSE continuously exceeded the threshold for

at least 0.5 seconds. The detected onset corresponded to the beginning of the first qualifying exceedance, and detection delay was calculated relative to the injected onset at 15 seconds.

5.3.3 Stress Prediction and Detection Results

Figure 7 compares the observed LAOS stress with the frozen t-ESSTV prediction. All four cases shared the same predicted stress because the imposed shear-rate histories and fitted parameters remained unchanged. As the perturbation magnitude increased, the deviation between observed and predicted stress became more pronounced after the injected onset.

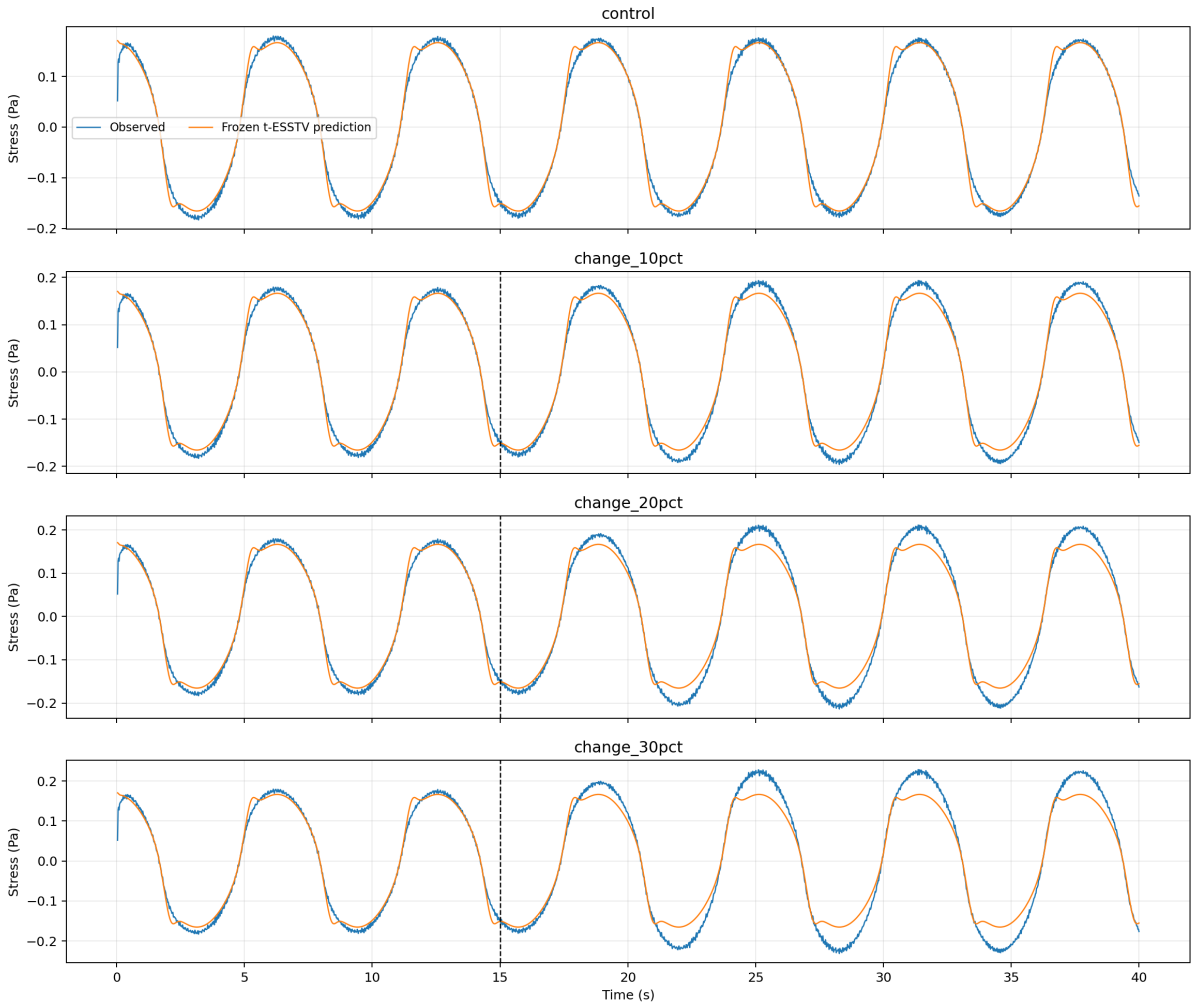


Figure 7: Observed APR18-based LAOS stress and frozen t-ESSTV predictions for the control and 10%, 20%, and 30% gradual stress-amplitude perturbations. Dashed vertical lines indicate the injected onset at 15 s.

Figure 8 presents the corresponding rolling RMSE values relative to the control-derived threshold. The unchanged control produced no alarm. The 10% perturbation did not trigger detection, whereas the 20% and 30% perturbations exceeded the threshold for the required duration. The larger perturbation was detected earlier.

Table 3 summarises the detection results. The 20% and 30% perturbations were detected with delays of 7.16 s and 4.44 s, respectively, while the 10% perturbation did not trigger an alarm.

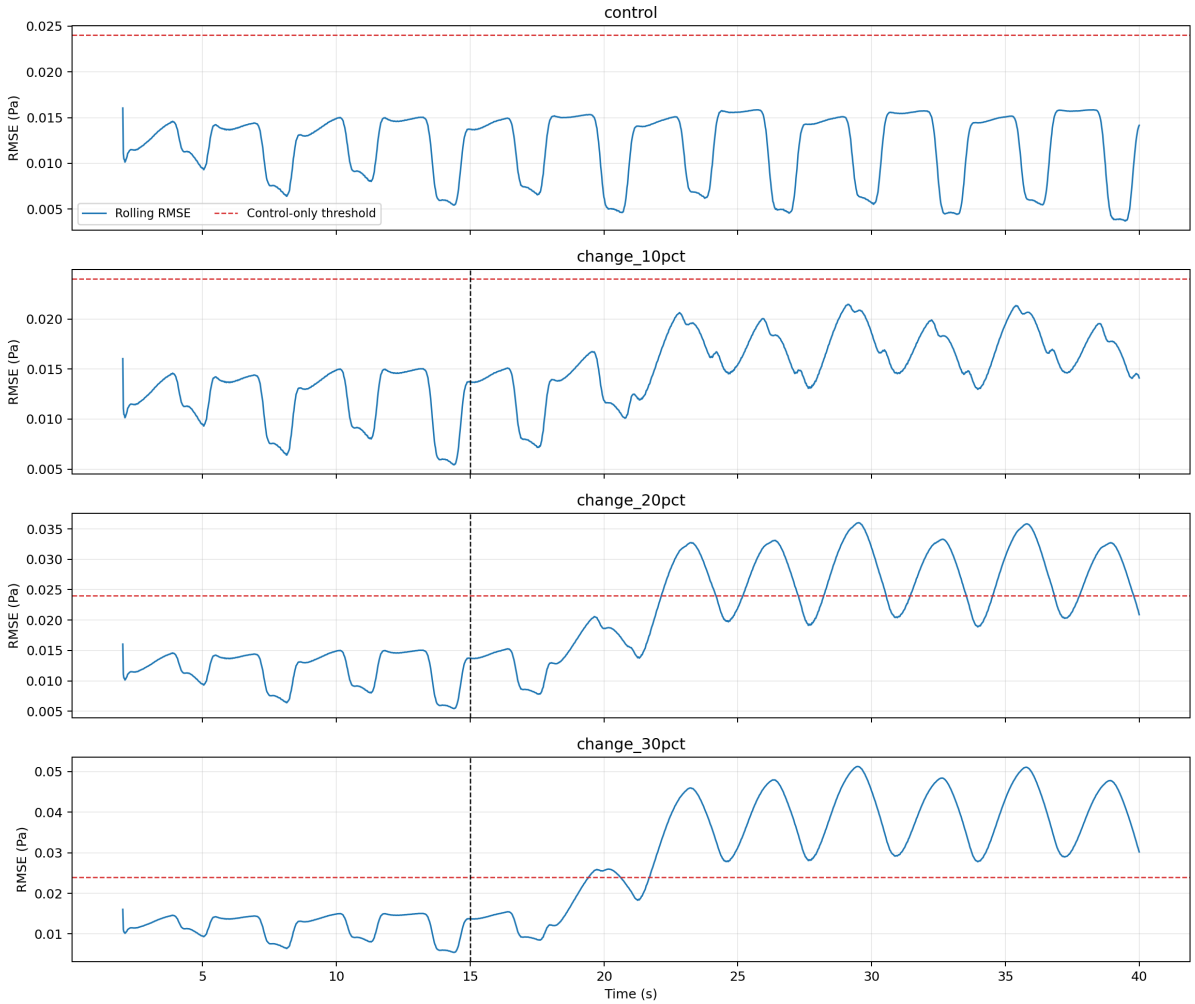


Figure 8: Two-second rolling RMSE for the control and perturbed LAOS records. Red dashed lines indicate the fixed detection threshold (0.0240178 Pa), while black dashed lines indicate the injected onset at 15 s. Detection required a continuous threshold exceedance of at least 0.5 s.

No pre-onset threshold exceedances were observed in the tested cases. These findings indicate greater sensitivity to larger stress-amplitude changes under the selected detection settings.

Table 3: Frozen t-ESSTV prediction and change-detection results for APR18-based semi-synthetic LAOS data. All injected changes begin at 15 s. A dash indicates no detected onset or a non-applicable value.

Case	RMSE (Pa)	Onset (s)	Delay (s)	Pre-onset samples
Control	0.01288	–	–	0
10% change	0.01522	–	–	0
20% change	0.02148	22.16	7.16	0
30% change	0.02924	19.44	4.44	0

5.4 Interpretation and Limitations

The calibration results showed excellent agreement with steady-state measurements, while transient stress predictions exhibited greater discrepancies. These limitations are relevant to residual-based change detection because model mismatch can contribute to the baseline prediction error.

The semi-synthetic experiments demonstrated that larger imposed stress changes could be distinguished from the unchanged control. However, gradual perturbations, temporal smoothing, and the persistence requirement contributed to detection delays. The detection threshold also limited sensitivity to smaller changes.

Several limitations should be considered. First, the benchmark changes were generated by scaling an observed LAOS stress record without modifying the corresponding strain or shear-rate histories. Therefore, the experiment evaluated sensitivity to imposed waveform deviations rather than a mechanistically simulated clotting process. Second, the threshold was defined using the same APR18 control waveform used in the benchmark, meaning that the absence of control alarms does not establish independent detection specificity. Third, residual prediction errors were present even under unchanged experimental conditions, so elevated RMSE cannot automatically be interpreted as a genuine rheological transition. Finally, the analysis considered only one baseline record, one change onset, and three perturbation magnitudes, limiting conclusions about generalisation across donors and experimental conditions.

Overall, t-ESSTV provides a physics-based reference for comparing measured stress with expected rheological behaviour. While its residuals can indicate departures from the calibrated response, they should be interpreted alongside conventional rheological measurements and signal-processing results. Independent experimental ground truth remains necessary to validate detection of genuine coagulation events.

6 Fast Fourier Transform and Short-Time Fourier Transform

6.1 FFT and STFT Analysis Methods

Whole-record FFT was used to examine the overall frequency content of strain and stress, while STFT tracked local stress features over time. FFT provided a spectral overview; STFT

supported time-resolved comparisons and reference-based confirmation. The primary settings are summarised in Table 4.

Table 4: Primary Fourier analysis and confirmation settings.

Setting	Value
Sampling rate	50 Hz
Nominal forcing frequency	$f_0 = 1/(2\pi) \approx 0.15915$ Hz
STFT window	628 samples; 12.56 s, approximately two cycles
Update interval	25 samples; 0.5 s
Window function	Periodic Hann
Harmonic orders	1, 3 and 5
Spectral feature band	$0.5f_0 \leq f \leq 8f_0$
Score threshold	$z \geq 5$ or $z \leq -5$
Persistence	Same-side exceedances spanning at least 1 s
Output timestamp	Exclusive observation-window end

6.1.1 Spectra and Feature Extraction

Each analysed segment was mean-centred and weighted by a periodic Hann window. Windowing reduces spectral leakage, although it also affects spectral separation[11]. Whole-record FFT results were expressed as one-sided peak-amplitude spectra with coherent-gain correction.

Harmonic amplitudes were evaluated directly at the nominal frequencies, rather than by selecting the nearest FFT bin:

$$I_h(m) = \frac{2}{\sum_{n=0}^{N-1} w[n]} \left| \sum_{n=0}^{N-1} w[n](x_m[n] - \bar{x}_m) e^{-\mathrm{i}2\pi h f_0 n/f_s} \right|, \quad h \in \{1, 3, 5\}. \tag{17}$$

Here, $x_m[n]$ is stress in window m, N is the sample count, f_s is the sampling rate, and $\bar{x}_m$ is the segment mean. The periodic Hann weights are $w[n] = 0.5 - 0.5\cos(2\pi n/N)$.

STFT applies Fourier analysis to successive windowed segments [2]. Eight features were extracted: I_1, I_3, I_5, I_3/I_1, I_5/I_1, spectral power, normalised spectral entropy and spectral bandwidth. Amplitudes are measured in Pa, while the ratios describe dimensionless relative harmonic contributions.

The three spectral statistics used the one-sided, window-normalised power spectral density within the selected band. With bin power $b_k = D_k \Delta f$, band power was $P = \sum_k b_k$ and normalised weights were $p_k = b_k/P$. Entropy was $H = -\sum_k p_k \log(p_k)/\log M$, where M is the number of included bins. Bandwidth was the power-weighted frequency standard deviation. Power, entropy and bandwidth have units of Pa^2, dimensionless and Hz, respectively.

Only complete windows were used, without padding. Outputs were assigned to exclusive window ends, representing when the complete observations became available. Consequently, the 0.5 s update interval does not represent the observation duration or detection delay.

6.1.2 Reference Scoring and Confirmation

Each feature was standardised separately using its reference median and median absolute deviation:

$$z_j(m) = \frac{F_j(m) - a_j}{s_j}, \qquad a_j = \mathrm{median}_{r\in\mathcal{R}}\, F_j(r), \qquad s_j = 1.4826\,\mathrm{median}_{r\in\mathcal{R}}\,|F_j(r) - a_j|. \quad (18)$$

The factor 1.4826 gives a normal-consistent dispersion estimate [16]. Invalid or nonpositive reference scales produced no valid score.

Synthetic references used all matched-control windows, making calibration offline. Original APR18 references used windows ending at or before 20 s; monitoring outputs ended after 20 s. Perturbed APR18 records reused the corresponding unmodified record's reference.

An alarm required same-side threshold exceedances spanning at least 1 s, equivalent to three consecutive outputs under the primary settings. A sign reversal, invalid score, time gap or below-threshold output interrupted the run. Confirmation time was the window end at which persistence was first satisfied. Synthetic evaluation retained only the first confirmation per feature, including any pre-onset confirmation, and used ground truth only after alarms were fixed.

6.2 Spectral Structure and Synthetic Feature Responses

Figure 9 compares the whole-record spectra of the common input strain and control stress. Strain was dominated by the fundamental, whereas stress also contained third- and fifth-harmonic components. Direct early-segment strain ratios were approximately 3.7×10^{-5} and 1.1×10^{-5}, compared with stress ratios of 0.100 and 0.0356. These results are consistent with a nonlinear stress response under an approximately sinusoidal input.

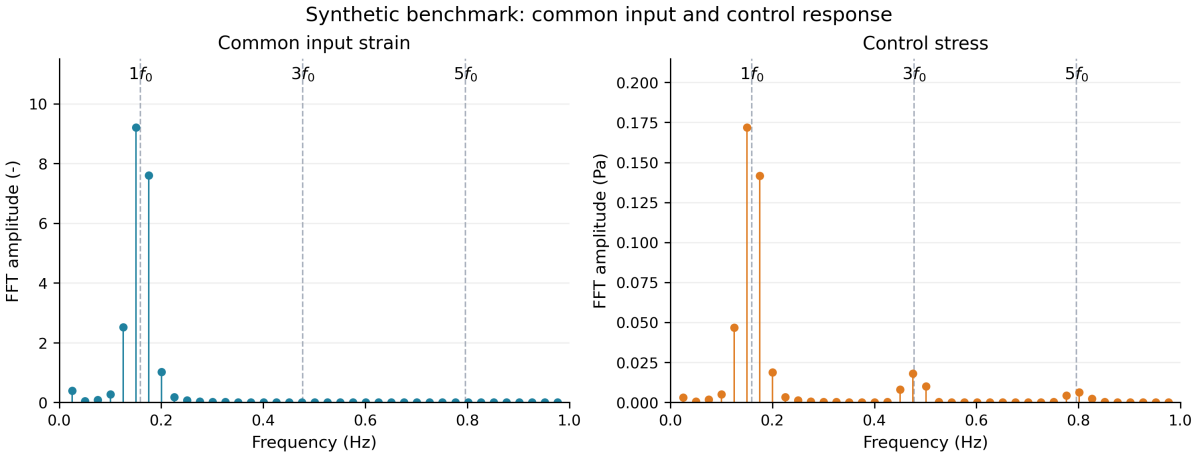


Figure 9: Whole-record FFT spectra of common input strain and control stress. Stems represent discrete FFT bins; dashed lines mark nominal harmonics. Panels retain their physical units and use separate amplitude scales.

Early and late harmonic estimates used $[2.00, 14.56)$ s and $[26.00, 38.56)$ s, respectively. Early estimates were identical across cases. Late absolute harmonic amplitudes increased by approximately 10%, 20% and 30% relative to the matched control, while relative changes in the harmonic

ratios remained below 0.002% in magnitude. The imposed gain therefore mainly scaled existing harmonics rather than increasing their relative contributions. Whole-record spectra alone did not locate the change onset.

Figure 10 shows selected STFT features as percentage differences from the matched control in the same window. These paired comparisons isolate perturbation responses from shared background variation; they are distinct from the reference-based scores.

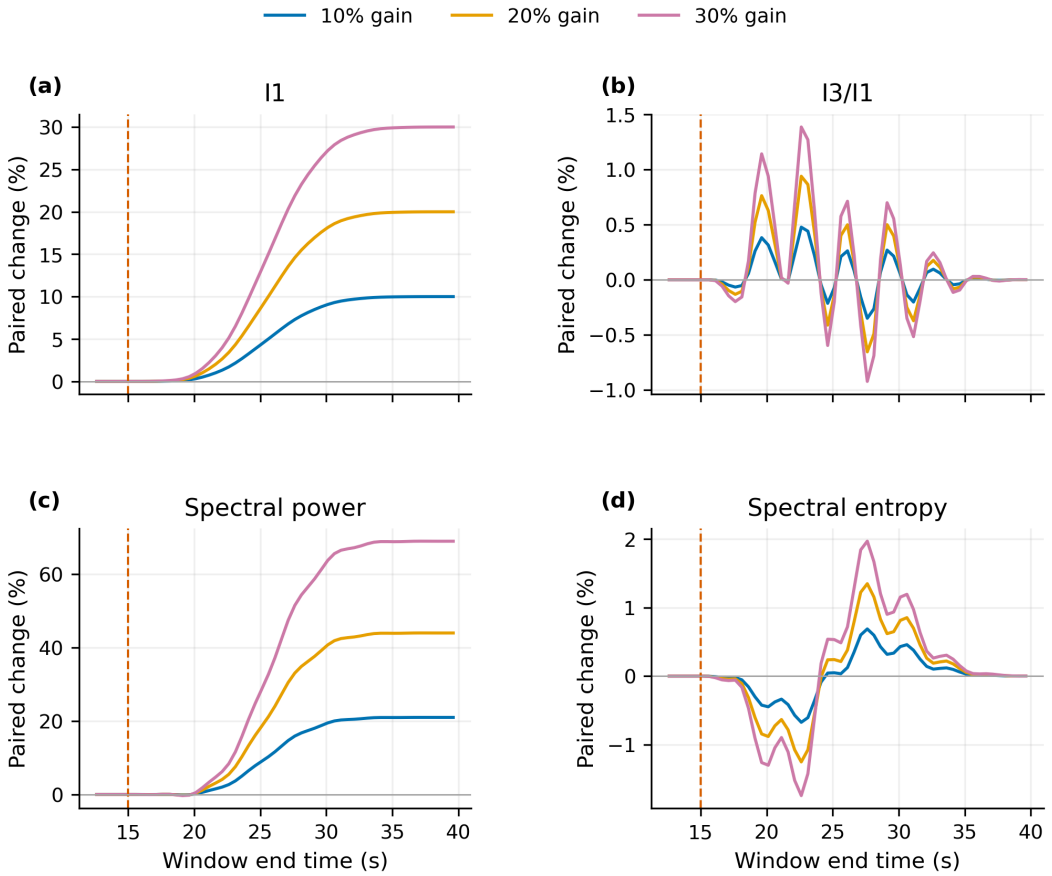


Figure 10: Selected STFT feature changes relative to the matched control in the same window. The dashed line marks the prescribed onset at 15 s. Outputs use window-end timestamps, and panels have separate vertical scales.

Late-window median harmonic increases approximately matched the imposed gains. Power increased by 21%, 44% and 69%, consistent with the squared-amplitude relationship $(1+\alpha)^2-1$. These summaries used three highly overlapping windows starting at or after 26 s, rather than independent repeated measurements.

Ratios, entropy and bandwidth were approximately unchanged once the gain settled, but could respond during the transition because scaling varied within an observation window. A temporary ratio or spectral-shape response therefore does not establish a persistent increase in material nonlinearity.

6.3 Synthetic Detection and Parameter Sensitivity

Table 5 reports confirmation delays relative to the prescribed 15 s onset. For the 10% case, I_1 and power confirmed first at 26.12 s. Entropy confirmed first for the 20% and 30% cases, with delays of 5.62 and 5.12 s. These earliest times are descriptive comparisons across features, rather than a validated fusion rule.

Table 5: First-confirmation delays in seconds under the primary two-cycle settings. ND denotes no confirmation within the record. No pre-onset confirmations occurred.

Feature	10% gain	20% gain	30% gain
I_1	11.12	9.12	8.12
I_3	14.62	10.62	8.62
I_5	ND	12.12	10.62
I_3/I_1	ND	ND	ND
I_5/I_1	ND	ND	ND
Spectral power	11.12	9.62	8.62
Spectral entropy	13.12	5.62	5.12
Spectral bandwidth	ND	ND	ND

Larger gains generally shortened amplitude-feature delays. Neither ratio nor bandwidth confirmed under the primary rule, consistent with their limited sustained response to uniform gain. Entropy's earlier confirmations reflected transition behaviour rather than a persistent late-stage spectral change. Feature-specific reference scales also influenced timing, so earlier confirmation did not necessarily mean a larger physical response.

No primary control alarms occurred. However, the control also supplied the reference, so this was an in-sample diagnostic rather than an independent false-alarm estimate.

6.3.1 Threshold and Persistence

Thresholds of 3, 5 and 8 were tested with persistence of 0, 1 and 3 s. At threshold 3 with 1 s persistence, three control features confirmed and nine gain-case–feature pairs confirmed before onset. Increasing persistence to 3 s removed these confirmations in this benchmark.

At threshold 5 without persistence, one control feature and three gain-case–feature pairs confirmed before the change. Adding the primary 1 s persistence removed these outcomes. Extending persistence to 3 s increased the earliest post-onset delays to 13.12, 7.62 and 7.12 s.

Timing must therefore be assessed alongside background and pre-onset alarms. Post-onset delay summaries exclude features whose first confirmation was pre-onset, so comparing those summaries alone can give a misleading parameter ranking.

6.3.2 Window Length

Each tested window length used its own matched-control reference. One-cycle windows (6.28 s) reduced I_1 delays to 8.34, 6.34 and 5.34 s. Improvements were feature-dependent: the 10% power delay increased to 12.34 s, and entropy no longer confirmed.

Stationary-mixture checks across 24 starting phases gave maximum relative errors below 0.05% for two-cycle harmonic amplitudes and ratios, and below 0.20% for one cycle. Half-

cycle errors were much larger, including approximately 485% for I_3, so half-cycle windows were excluded from the main comparison. These checks concern the tested stationary mixture, not detection accuracy.

Shorter windows can improve availability while altering feature estimation and reference variation. Entropy and bandwidth also use different frequency grids at different lengths. The one-cycle window remains longer than 5 s; reliable operation below that observation duration was not demonstrated.

6.4 Original APR18 Feature Variation and Reference Departures

Figure 11 shows selected features relative to each record's first complete window. The six curves represent separate imposed strain-amplitude conditions, not successive stages of one experiment. The first window is a plotting origin, distinct from the scoring reference.

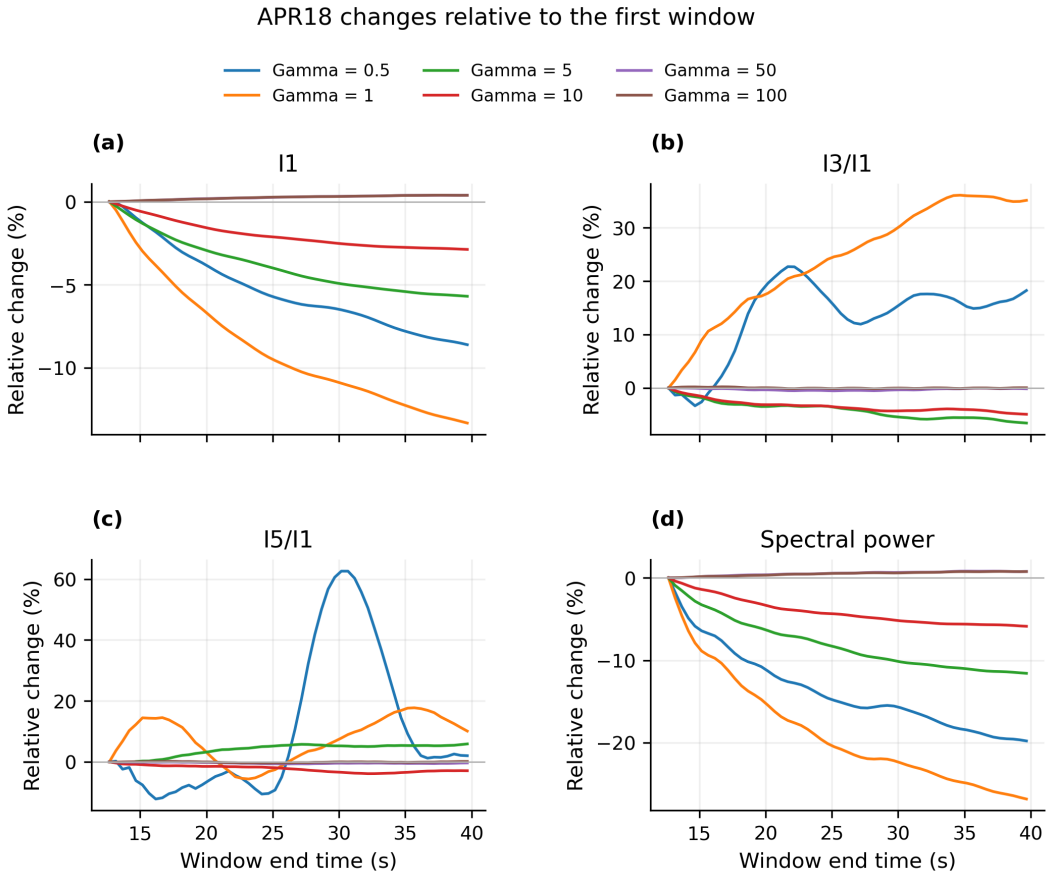


Figure 11: APR18 feature changes relative to each record's first complete window. Curves represent separate strain-amplitude conditions. The plotting origin is not a verified stable state. Outputs use window-end timestamps.

I_1 and power declined at amplitudes of 0.5, 1, 5 and 10, with the largest decline at 1. Both changed relatively little at 50 and 100. Ratios followed different patterns: I_3/I_1 generally increased at 0.5 and 1, while I_5/I_1 showed a pronounced temporary increase at 0.5. The original records therefore contained variation beyond uniform stress scaling.

Input checks showed fundamental-amplitude ranges below 0.009% of nominal amplitude

across all six records, with very small input higher harmonics. This supports examining stress-feature variation, without establishing its physiological cause.

Under the primary rule, I_5 and I_5/I_1 confirmed upward departures at 28.18 s for amplitude 0.5. Downward ratio departures occurred at 29.18 s and 22.18 s for amplitudes 10 and 50. Additional upward confirmations occurred in I_3 at 36.18 s for amplitude 50 and power at 38.18 s for amplitude 100. No features confirmed at amplitudes 1 or 5.

Visible percentage changes did not directly determine alarms because scoring used each feature's reference distribution. Conversely, a large score could represent a small change: the amplitude-50 ratio confirmation had $z \approx -8.50$, despite being only 0.26% below its reference median.

Reference endpoints of 16, 18 and 20 s were compared with monitoring held fixed. Some outcomes were consistent, including the amplitude-50 ratio confirmation at 22.18 s. Others changed substantially: amplitude-100 power confirmation shifted from 25.18 s with earlier endpoints to 38.18 s under the primary reference.

The primary reference contained 15 highly overlapping outputs, with only one complete non-overlapping window fitting within its interval. Early monitoring windows also overlapped the reference period. These constraints limit reference reliability and temporal interpretation. Without verified event labels, confirmation times describe reference departures rather than physiological change times.

6.5 Controlled Perturbations and Noise Sensitivity

Supplementary APR18 experiments introduced gain steps, third-harmonic additions and 5 s noise bursts at 20 s. Strengths were 5%, 10% and 20%: gain fractions for multiplicative changes, and added RMS relative to pre-onset centred stress RMS for additive changes. Ten noise seeds were tested per condition and strength. These experiments were separate from the gradual-gain benchmark.

Fully post-step windows reproduced expected scaling: harmonic amplitudes increased by 5%, 10% and 20%, and power by 10.25%, 21% and 44%. Ratios and spectral shape remained unchanged, although transition windows could respond. Third-harmonic additions produced sustained changes in I_3, I_3/I_1 and spectral-shape features, with responses depending on the original magnitude and phase.

Figure 12 compares first confirmations with each unmodified counterpart. This distinguishes newly introduced alarms from earlier, unchanged, later or removed background confirmations.

Gain steps introduced new alarms in 11 of 18 versions for I_1 and 9 of 18 for power. Third-harmonic additions introduced new alarms in all 18 versions for I_3/I_1, entropy and bandwidth. However, noise introduced new alarms in 89, 112 and 146 of 180 versions for these same features; I_5 produced 129 new alarms. Sensitivity to a controlled change therefore did not establish specificity to that change.

The persistence rule did not eliminate noise confirmations. Highly overlapping windows retained the same affected samples across consecutive outputs, so those exceedances were not independent evidence. Stricter settings reduced some confirmations: total entropy alarms in noise versions decreased from 112/180 under primary settings to 72/180 at threshold 8 with 3 s

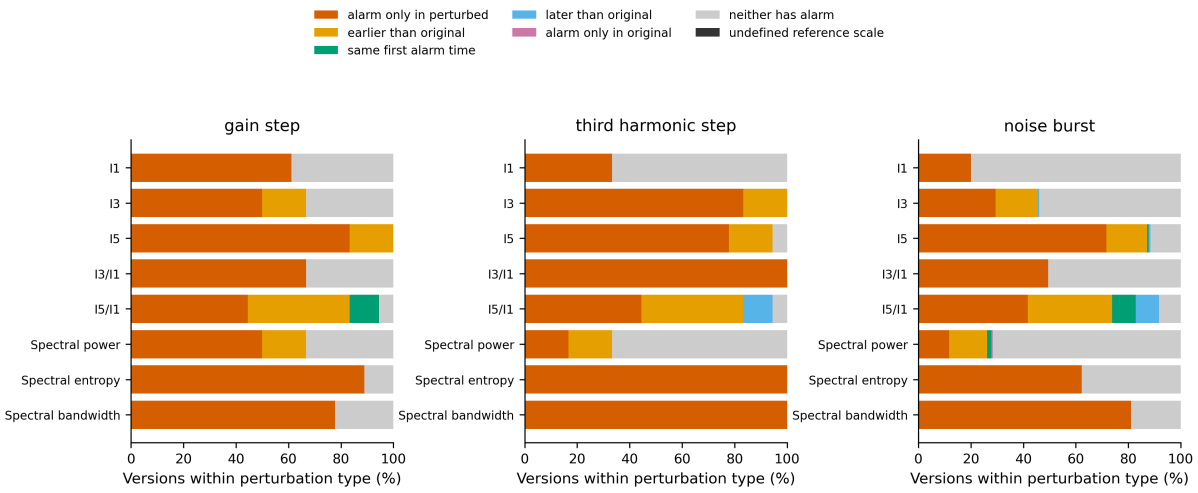


Figure 12: Paired first-confirmation outcomes relative to unmodified APR18 records. Each feature has 18 gain versions, 18 third-harmonic versions and 180 noise versions. Pooled proportions are not independent donor rates.

persistence. Such totals include existing background departures and require paired interpretation.

After a complete window no longer contained noise-affected samples, paired feature differences returned to zero. This follows from finite-window analysis and unchanged post-burst samples, rather than demonstrating successful noise rejection. Repeated seeds quantify simulation variability on shared backgrounds, not uncertainty across donors.

Conditional lag was calculated only when confirmation occurred. It cannot establish detection success alone because an alarm may reflect background variation and unconfirmed versions are excluded. Monitoring began after injection, so pre-injection alarms were not evaluated in this protocol.

6.6 Summary and Limitations

FFT described overall harmonic structure, while STFT tracked local changes and supported reference-based confirmation. Absolute amplitudes and power responded clearly to gain; ratios helped distinguish scaling from relative harmonic changes. Confirmation depended on feature, window, threshold and reference selection. Noise-associated alarms and unlabelled APR18 departures prevent physiological interpretation. Independent labelled records are needed to evaluate missed changes, background alarms and delay jointly. Complete feature definitions, numerical checks, parameter studies and event intervals remain available in the accompanying notebooks and exported outputs.

7 Continuous Wavelet Transform (CWT)

The CWT analysis examines whether local time–frequency features can reveal changes in large amplitude oscillatory shear (LAOS) stress signals. Wavelet analysis provides frequency-dependent localisation, which motivates its use alongside fixed-window Fourier analysis [10, 20]. The

experiment is divided into two main parts: first, determining whether CWT responds to controlled variations in stress-gain and comparing its performance with that of STFT; and second, assessing its ability to characterize the harmonic structure present in the raw APR18 measurement data.

7.1 CWT Method and Analysis Settings

The implementation uses explicitly sampled complex Morlet kernels centred within a finite observation window. The mother wavelet is

$$\psi(u) = \frac{1}{\sqrt{\pi B}} \exp(-u^2/B) \exp(2\pi \mathrm{i} u), \qquad B = 3. \tag{19}$$

Its nominal centre frequency is one, giving a discrete scale of $s = f_s/f$ at analysis frequency f. Each sampled conjugate kernel has its discrete mean removed. This is equivalent to removing the window mean before projection. All coefficients use observations inside the current buffer; the finite buffer nevertheless truncates the wavelet, especially at low frequencies. Discrete amplitude calibration is therefore required.

The nominal forcing frequency is $\omega_0 = 1$ rad s^{-1}, so $f_0 = \omega_0/(2\pi) \approx 0.15915$ Hz. Table 6 summarises the settings. Two oscillation cycles form the primary window, while one cycle is used for sensitivity analysis. Each window length has a separately calibrated control reference.

Table 6: Settings used in the finite-window CWT analysis.

Setting	Value or definition
Uniform sampling rate	50 Hz, after interpolation
Wavelet	Complex Morlet, $B = 3$, nominal centre frequency 1
Primary window	628 samples, 12.56 s, approximately two cycles
Sensitivity window	314 samples, 6.28 s, approximately one cycle
Output interval	0.5 s
Harmonic frequencies	f_0, $3f_0$, and $5f_0$
Spectral feature band	$0.5f_0 \le f \le 8f_0$, sampled on the window's rFFT frequency grid
Primary alarm	Same-side robust score of at least 5 in magnitude for 1 s
Alarm timestamp	Exclusive end of the confirming observation window

The wavelet is centred for estimation, but its output becomes available only after the complete window has been observed. Detection times therefore use the window end, rather than its centre or the first threshold crossing. This avoids backdating a confirmed event. It does not make the complete pipeline prospective, because the benchmark reference uses the full matched-control record.

7.2 Time–Frequency Feature Extraction

Eight features are extracted from each window: the harmonic amplitudes I_1, I_3, and I_5; the ratios $R_3 = I_3/I_1$ and $R_5 = I_5/I_1$; a band-power proxy; normalised spectral entropy; and spectral bandwidth. Absolute amplitudes describe response magnitude, whereas ratios describe

relative harmonic contributions. Retaining both helps distinguish an approximately uniform gain from a change in harmonic composition.

For window m, let $C_m(f) = \sum_n x_m[n]k_f[n]$, where k_f is the sampled, mean-removed conjugate kernel. Harmonic amplitudes are estimated as

$$I_h(m) = \frac{2|C_m(hf_0)|}{G(hf_0)}, \qquad h \in \{1, 3, 5\}, \tag{20}$$

where $G(f)$ is the magnitude of the kernel's discrete positive-frequency response. This corrects the amplitude gain but does not eliminate negative-frequency interference or leakage from other harmonics.

At selected band frequencies f_j, define

$$D_j = \frac{2|C_m(f_j)|^2}{f_s \sum_n |k_{f_j}[n]|^2}, \qquad P = \sum_{j=1}^{M} D_j \Delta f, \qquad p_j = \frac{D_j \Delta f}{P}, \tag{21}$$

$$H = -\frac{\sum_{j=1}^{M} p_j \log p_j}{\log M}, \qquad \bar{f} = \sum_{j=1}^{M} p_j f_j, \qquad \mathrm{BW} = \sqrt{\sum_{j=1}^{M} p_j (f_j - \bar{f})^2}. \tag{22}$$

Here, $\Delta f = f_s/N$, and zero-probability entropy terms contribute zero. P has units of Pa^2, H is dimensionless, and BW is measured in hertz. P is an integrated wavelet-density proxy, not an identical estimator of the Hann-window Fourier power used by STFT. Consequently, raw spectral-feature values should not be treated as interchangeable between methods.

7.3 Controlled Gain-Change Experiments

Figure 13 shows the feature responses in synthetic data. In the late window $[26.00, 38.56)$ s, the harmonic amplitudes increase by approximately the imposed gains. The power proxy increases by 20.99%, 43.98%, and 68.97%, close to the constant-gain prediction $(1+\alpha)^2-1$. Both harmonic ratios change by less than 0.008% relative to their matched-control values in this window.

These results distinguish signal magnitude from relative harmonic structure. A common multiplicative gain increases absolute harmonics while approximately preserving their ratios once the gain settles. Entropy and bandwidth can still change during the transition because the gain varies within a window. Such responses do not, by themselves, demonstrate increased material nonlinearity.

7.4 Change-Detection Performance and Feature-Specific Trade-offs

Each feature F_j is standardised using the median m_j and median absolute deviation of all 55 primary control outputs:

$$z_j(t) = \frac{F_j(t) - m_j}{1.4826\,\mathrm{MAD}_j}. \tag{23}$$

Invalid or zero reference scales produce no valid score. The primary rule requires $z_j \geq 5$ or $z_j \leq -5$ on the same side for at least 1 s, corresponding to three adjacent outputs. A sign reversal, invalid score, or missing output breaks the run. Scores and alarms are computed before onset labels are used for evaluation; the delay is the confirmation time minus 15 s.

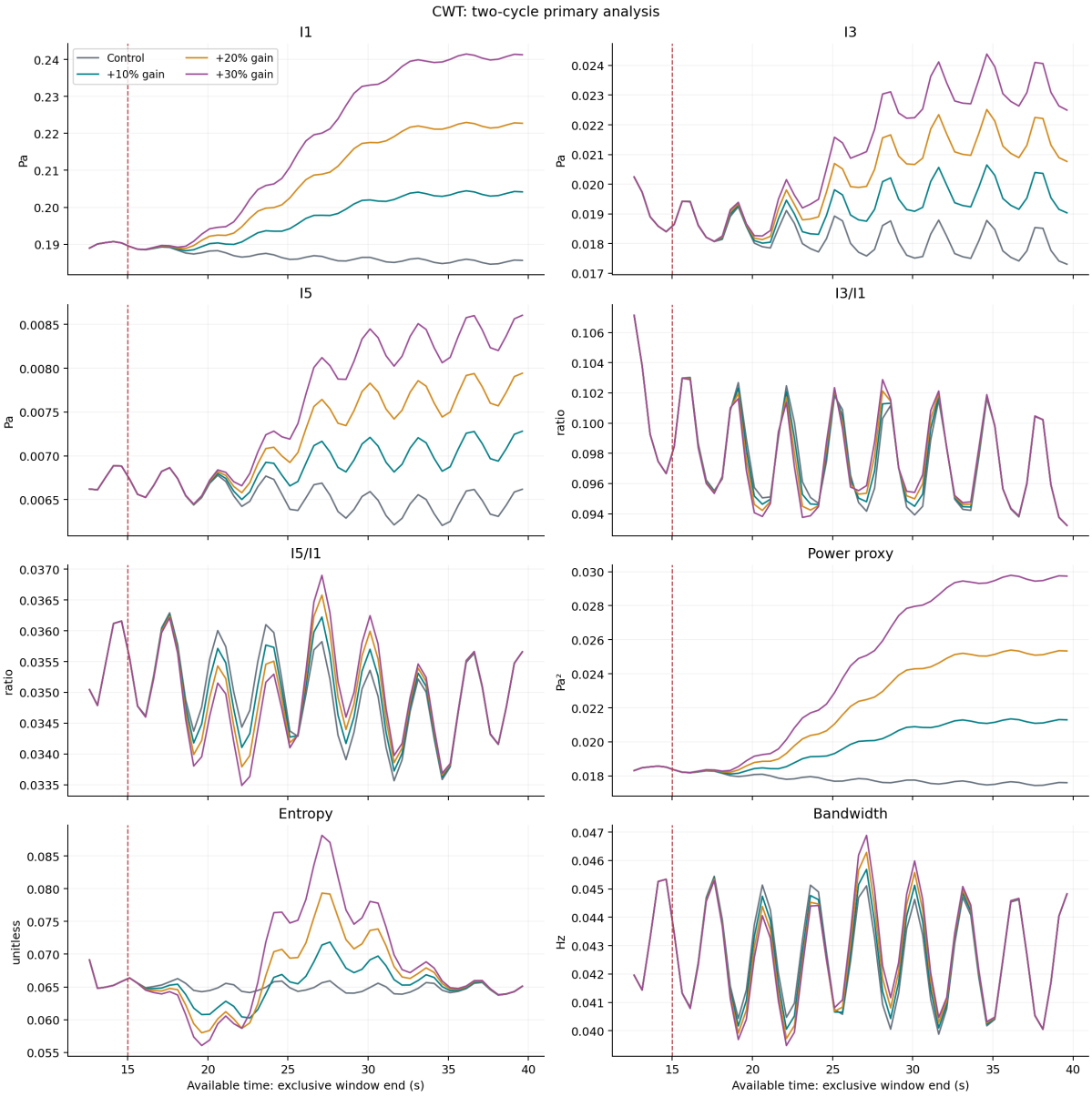


Figure 13: Two-cycle CWT features for the matched control and three gain cases. The dashed line marks the imposed onset at 15 s. The horizontal axis uses the exclusive window end. Absolute amplitudes and power separate as gain increases, while harmonic ratios remain close to the control.

The primary CWT I_1 delays are 11.12, 8.12, and 6.12 s for the three increasing gains. Power gives similar delays of 11.12, 8.12, and 6.62 s. I_3 and I_5 confirm only the 20% and 30% cases. Neither harmonic ratio nor bandwidth confirms a change under the primary rule. Their absence of alarms is consistent with limited sensitivity to this particular gain perturbation, rather than evidence that these features are generally uninformative.

No primary control alarm or pre-onset confirmation occurs in these CWT outputs. However, the control also supplies the reference, so this is an in-reference diagnostic, not an independent false-alarm estimate. The three gain cases share one background, and overlapping windows are not independent observations. Accordingly, these results do not establish a population detection probability or a general minimum detectable gain.

Window length introduces a further trade-off. One-cycle windows reduce the I_1 delays to 8.34, 5.84, and 5.34 s, but harmonic-mixture calibration reveals substantially greater high-order estimation error. For a known mixture with amplitudes 1, 0.10, and 0.03, tested at 24 starting phases, the maximum two-cycle errors are 0.43%, 4.09%, and 2.94% for I_1, I_3, and I_5. With one cycle, the latter two rise to 28.62% and 25.25%. Faster confirmation therefore does not imply more accurate harmonic estimation; the two-cycle setting remains the primary comparison.

7.5 Comparison with STFT

The STFT comparison uses the team's reported results on the same four benchmark signals. Both analyses use 50 Hz sampling, a two-cycle observation window, a 0.5 s output interval, the same eight feature categories, full-control reference calibration, and the same persistence rule. STFT uses periodic Hann weighting; CWT uses frequency-dependent Morlet weighting. The numerical feature definitions therefore remain method-specific despite the matched evaluation conditions.

Table 7: First confirmation delays relative to the 15 s injection onset, in seconds. Primary two-cycle results; smaller delays indicate earlier confirmation when an alarm occurs.

	10% gain		20% gain		30% gain	
Feature	CWT	STFT	CWT	STFT	CWT	STFT
I_1	11.12	11.12	8.12	9.12	6.12	8.12
I_3	ND	14.62	17.12	10.62	13.62	8.62
I_5	ND	ND	12.62	12.12	12.12	10.62
I_3/I_1	ND	ND	ND	ND	ND	ND
I_5/I_1	ND	ND	ND	ND	ND	ND
Power	11.12	11.12	8.12	9.62	6.62	8.62
Entropy	13.12	13.12	5.12	5.62	5.12	5.12
Bandwidth	ND	ND	ND	ND	ND	ND

Note. ND denotes no confirmed alarm by the final available window under the primary rule; it is not a zero delay. STFT values are taken from the team's FFT/STFT progress report and cross-checked against the supplied comparison table.

Table 7 shows a feature-dependent outcome. For I_1, CWT matches STFT at 10% gain and confirms 1 and 2 s earlier at 20% and 30%. Its power feature also confirms earlier in the larger-gain cases. In contrast, STFT detects the 10% change through I_3, which CWT does not, and

confirms the larger I_3 changes 6.5 and 5 s earlier. STFT is also earlier for I_5 when both methods confirm. Some differences are only one 0.5 s output interval and should not be overinterpreted.

Entropy confirms relatively early in the larger-gain cases, but this measures a departure in normalised spectral distribution, not a specific physiological mechanism. Likewise, a clearer scalogram does not necessarily imply better detection timing. The comparison separates three criteria: feature estimation accuracy, confirmed-event timing, and visual interpretability. The present evidence supports complementary strengths rather than a general ranking of CWT above STFT.

7.6 CWT Analysis of the Original APR18 Data

The original-data analysis covers six separate APR18 LAOS records at $\omega_0 = 1\ \mathrm{rad\,s^{-1}}$, using the supplied γ labels 0.5, 1, 5, 10, 50, and 100. The fixed startup sample is removed from each block before interpolation. The two-cycle analysis gives 55 overlapping windows per condition.

Figure 14 shows a persistent fundamental band, a third-harmonic band, and a weaker fifth-harmonic contribution. Each panel is normalised to its own maximum, so colour intensity cannot establish absolute amplitude differences between conditions. The scalogram uses window-centre timestamps for visualisation; these are distinct from the availability timestamps used for detection.

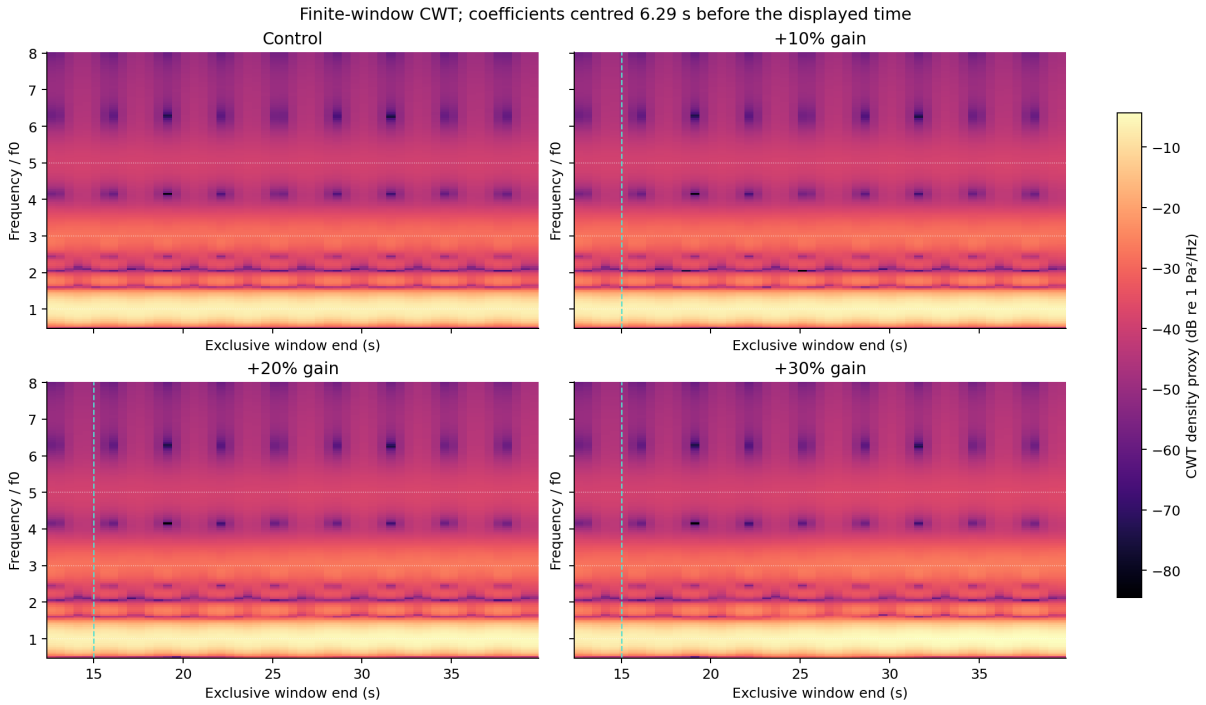


Figure 14: Two-cycle CWT scalograms for the six original APR18 conditions. Frequency is expressed relative to f_0, with reference lines at the first, third, and fifth harmonics. Power is shown in decibels relative to each panel's own maximum. The horizontal axis denotes the window centre.

Table 8 summarises median features. I_1 increases across the supplied conditions, whereas the harmonic ratios are non-monotonic: the largest median R_3 occurs at $\gamma = 10$, and the largest R_5

at $\gamma = 50$. These are comparisons across separately imposed conditions, not successive stages of a clotting trajectory.

Table 8: Two-cycle CWT summaries of the original APR18 records. Ratios and amplitudes are medians over windows.

APR18 γ label	I_1 (Pa)	I_3/I_1	I_5/I_1	I_1 robust CV (%)
0.5	0.02787	0.04172	0.00502	2.73
1	0.04239	0.07509	0.01320	4.21
5	0.12213	0.08848	0.02609	1.87
10	0.18644	0.09707	0.03501	0.81
50	0.53727	0.08303	0.03716	0.33
100	0.89172	0.06922	0.03177	0.34

Note. Robust CV is $100 \times 1.4826\,\mathrm{MAD}/|\mathrm{median}|$. It describes within-record variation, not uncertainty across independent samples. The γ column retains the condition labels in the supplied result files.

A same-window harmonic regression provides a further method-agreement check. Across the six records, the maximum absolute difference between CWT and regression median amplitudes is 0.163% for I_1 and 1.427% for I_3. The weak I_5 component at $\gamma = 0.5$ differs by 17.386%, illustrating the difficulty of estimating small harmonics. Agreement of record-level medians does not guarantee agreement in every individual window, and regression is a comparison estimator rather than independent ground truth.

No labelled clotting or mechanical-change time is supplied for these records, so detection accuracy and delay cannot be calculated. Rheometrical gel-point studies associate incipient clot formation with specific viscoelastic criteria [7, 6]; the present harmonic summaries do not independently establish those criteria. These results support descriptive characterisation and estimator checks, while physiological event identification requires additional validation with other model and methods.

7.7 CWT Results Summary

CWT provides interpretable local features that distinguish absolute stress magnitude from relative harmonic composition. In the controlled benchmark, I_1 and power identify all three tested gain changes, while stable ratios help explain the nature of the perturbation. CWT confirms some larger-gain changes earlier than STFT, but STFT performs better for several higher-harmonic detections. The original APR18 records further demonstrate useful harmonic summaries alongside greater uncertainty for weak components.

Three priorities follow for subsequent method development: improve short-window harmonic separation; evaluate reference and alarm settings on independent recordings; and test robustness to changes beyond uniform gain, including harmonic redistribution and noise. Any multi-feature detector should be defined and calibrated before evaluating independent cases, rather than selecting the earliest feature retrospectively. Linking these mechanical indicators to clot formation will additionally require matched physiological or rheological reference measurements.

8 Discrete Wavelet Transform (DWT)

8.1 Aim, Data and Method

We investigated DWT to separate stress responses by time scale and extract features for change-point detection. We analysed real APR18 STEP and LAOS records and used synthetic data for controlled validation.

Why synthetic data were used. APR18 has no independently labelled material-change onset, clean stress trace or separate noise record. Neither a loading step nor a visible peak provides ground truth for material change. Synthetic data supply known onsets, clean responses and no-change controls for testing features and detection rules. Real records assess interpretation and sampling sensitivity, but cannot establish detection accuracy or physiological validity. Synthetic references were supplied with the data packages.

Table 9: Selected data and their purpose. Synthetic records are scenarios, not donors.

Data	Selection and purpose
Synthetic STEP	`control_1/test_clot_1`, 100 Hz, 120 s; loading fixed after 0.05 s, extra growth prescribed at 40 s.
Synthetic oscillation	Initial four-record clean/noise study; separate LAOS `control/clot_A` onset comparison (300 s). Both: 100 Hz, 30 min, 1 Hz loading.
APR18 STEP	0.1 to 1 s^{-1}, BH–BK, 65 nonuniform points over approximately 63 s; full recorded rise and relaxation.
APR18 LAOS	$\omega = 1$ rad/s, $\Gamma = 10$, M–O, 1,996 points; periodic/harmonic interpretation. Separate $\Gamma = 5$ background-sensitivity study.

Processing and features. DWT uses filtering and downsampling to separate resolutions [13]. Reconstructed components satisfy $\sigma = A_J + \sum_{j=1}^{J} D_j$: A_J contains slower structure and D_j describes activity in one band. Change-analysis applications support task-dependent scale selection, not a universal best level [19]. For each window, we used:

- $M_j = \text{mean}(A_j)$ and its between-window rate: trend, direction and speed of change; ordinary mean rate was the STEP baseline.
- $E_j = \text{mean}(D_j^2)$ and $R_j = \sqrt{E_j}$: scale-specific strength; ordinary stress RMS was the LAOS baseline. $H_{13} = \sqrt{E_1 + E_2 + E_3}$ and $P_6 = E_6 / \sum_1^8 E_j$ described fast activity and relative detail share, not noise labels.

All analyses were offline, without amplitude normalisation or threshold denoising. Synthetic data needed no interpolation. All real STEP points anchored linear interpolation onto 201,601 uniform points ($\Delta t \approx 0.3125$ ms); grid refinement checked numerical stability. LAOS used five cycles, 7–38.416 s, on 1,536 uniform points. Interpolation creates no new measurements. STEP windows were 1 s (slow) or 10 ms (early); LAOS onset windows were 10 s, advanced by 1 s. Rates use the later window centre; overlapping windows and future filtering prevent a real-time interpretation.

8.2 Synthetic Results: Growth, Noise and Onset

With db2, symmetric extension and eight-level decomposition, A_8 rate tracked prescribed STEP growth (Figure 15). Its mean over 40–70 s was 6.046×10^{-4} Pa/s, close to the clean value 6.083×10^{-4} Pa/s; control remained near zero. Ordinary mean rate also captured growth. Lower background fluctuation does not establish better detection because smoothing differs.

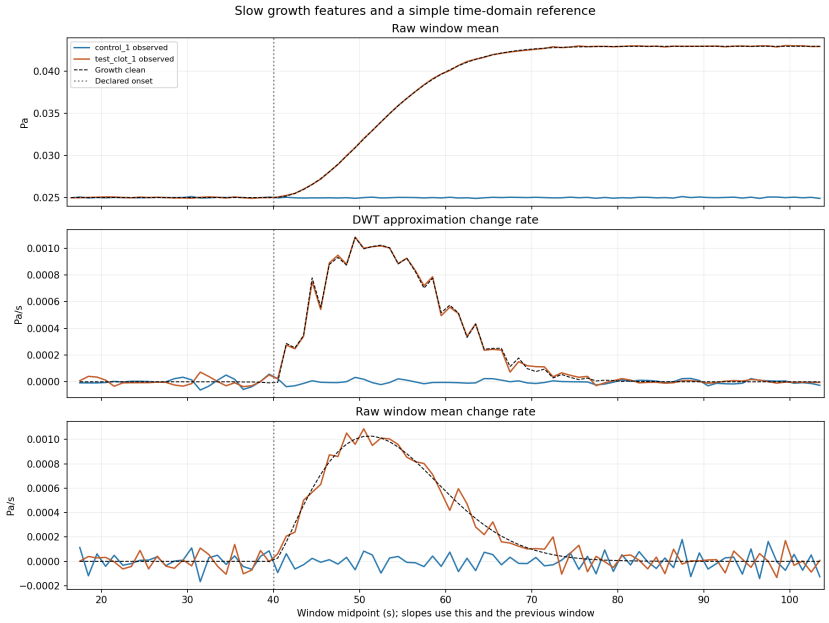


Figure 15: Synthetic STEP: mean stress, A_8 rate and ordinary mean rate. The 40 s line is the prescribed onset, not a detection. Both rates reveal growth; the largest peak occurs later than onset.

Fast strengths R_1–R_3 poorly distinguished slow growth from control; R_8 responded but was alignment-sensitive. Increasing levels 7–9 reduced observed-versus-clean rate error but increased clean-trend distortion. More smoothing did not necessarily improve onset timing.

Noise controls changed interpretation. In the initial oscillation study, D1 shifted from noise-dominated to clean-response-dominated as the response grew. The constant-material, noise-free control still showed 4.79% peak-to-peak variation in R_6 with an 80 s repeat due to downsampling/window alignment. Noise could reverse P_6 change. Fast details cannot be treated as noise by default.

Table 10: Separate synthetic LAOS pair: onset prescribed at 300 s. Threshold: control median $+6 \times 1.4826\,\mathrm{MAD}$ (20–200 s), three consecutive windows.

Result	DWT R_6	Ordinary RMS
Estimated onset / signed error (s)	302 / +2	298 / −2
End of third qualifying window (s)	309	305
Control candidate intervals	0	0

Both features had the same absolute timing error. The earlier RMS estimate used future samples within its window; it was not a forecast. One control and one positive record cannot establish general accuracy. DWT offered no localisation advantage here.

8.3 Real APR18 STEP: Trend and Sampling Limits

Measured stress reached a sampled minimum of 0.006680 Pa at 0.007876 s, rose to 0.05972 Pa at 0.450381 s, then fell to 0.049528 Pa at the final observation (17.1% below the peak). These extrema do not establish material-change events. From 0.042 to 1 s, measured shear rate remained near 1 s^{-1} while stress continued to change, so input and output did not settle together.

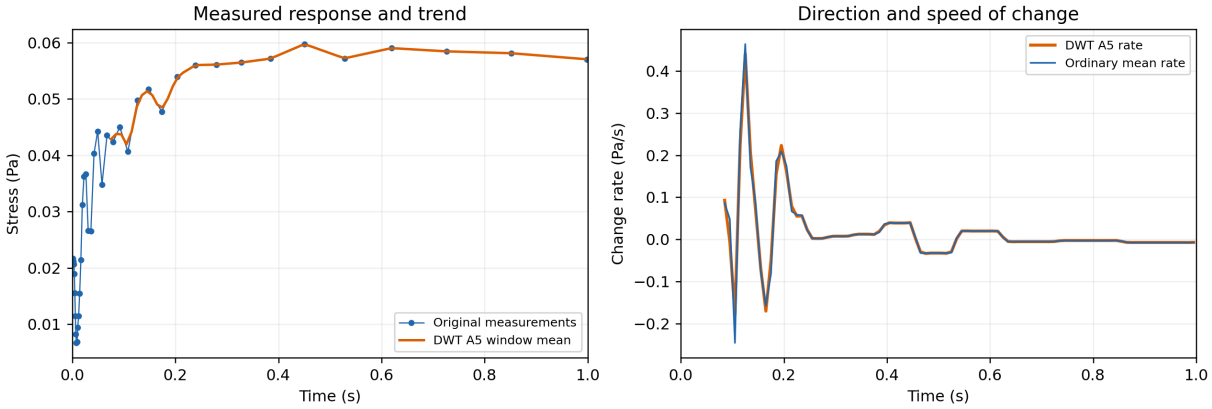


Figure 16: Early STEP response: original measurements and A_5 mean (left), A_5 and ordinary rates (right). Rate curves begin at 0.085 s after boundary screening, so the earliest valley is outside this validated range.

The A_5 rate describes the direction and speed of stress change. It closely follows ordinary mean rate, supporting response description but not improved timing accuracy. At later times, A_{13} captured a broad decline, with a mean rate of -1.738×10^{-5} Pa/s across 28 eligible differences in 18–47 s. We analysed early and slow scales separately because boundary exclusion at the slow scale would remove the early response.

Table 11: Checks distinguish numerical stability from observational support. Relative differences below concern feature vectors, not material-change percentages.

Check	Result and implication
Sampling support	43 of 62 one-second windows contain no original point. Fine interpolation cannot recover unmeasured dynamics.
Four-fold grid refinement	Early A_5 rate differed by 2.20%: reasonable convergence for the assumed linear interpolant.
Analysis start shifted 5 ms	A_5 rate differed by 14.9%; R_6/R_7 by approximately 47–48%. Individual peaks are alignment-sensitive.
Synthetic trace at APR18 times	The known 40 s onset fell in the 39.0485–45.7985 s gap; interpolation spread the rise across this interval.

The boundary screen excluded 6.25 scale spacings at each end (approximately 16 s for the slow scale). This practical rule is not a rigorous uncertainty bound. The checks support describing real-data changes as trends or candidate intervals, rather than precisely located material-change points. Differences from synthetic results also reflect sampling and loading protocol, not only signal behaviour.

8.4 Real APR18 LAOS: Periodic Components and Fast Activity

For $\Gamma = 10$, we applied db4 with periodic extension and total depth $J = 8$. We fitted a constant and harmonics 1, 3 and 5 by least squares, then subtracted the constant and fundamental to obtain the residual. We reconstructed the D6 component of both this residual and the fitted third harmonic.

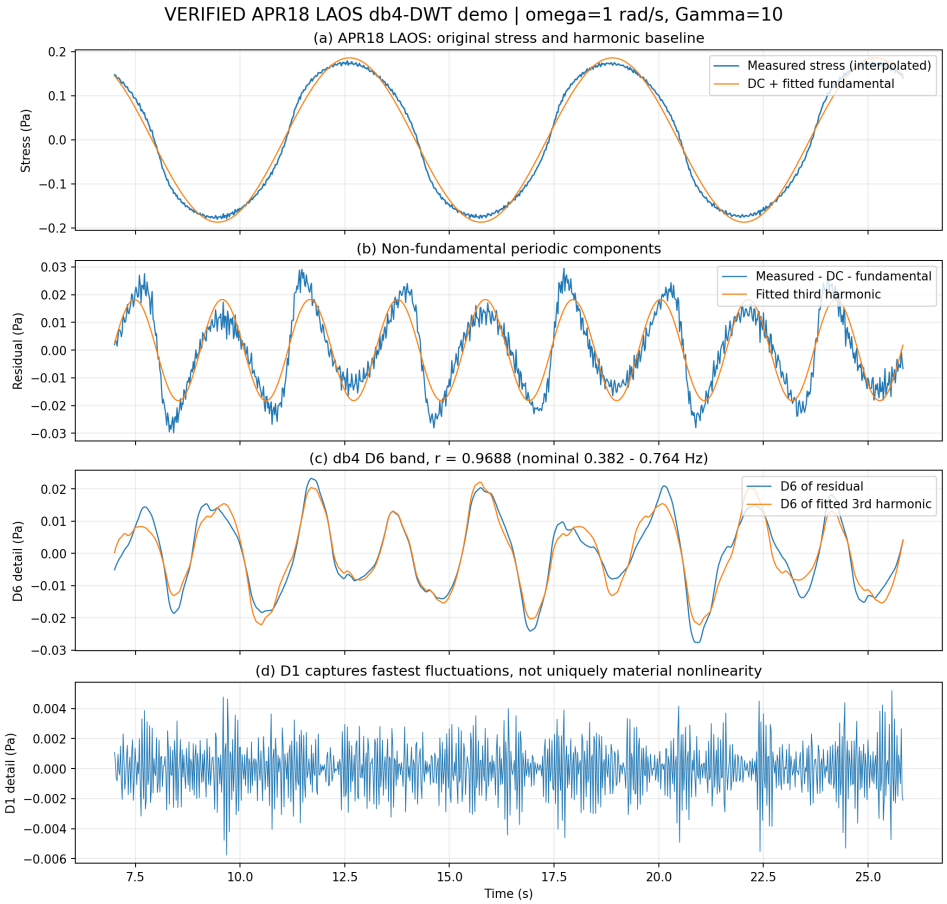


Figure 17: (a) Stress and fundamental baseline; (b) residual and fitted third harmonic; (c) their D6 components; (d) D1 of original interpolated stress. Three cycles are displayed; the correlation uses all five analysed cycles.

The fit at the original timestamps gave $I_3/I_1 = 9.79\%$. The D6 curves had a correlation of $r = 0.9688$, consistent with a repeating third-harmonic contribution at this scale. Because the reference was fitted to the same data, this correlation measures neither detection accuracy nor complete separation of harmonics from noise. The causes of the faster D1 fluctuations remain unresolved.

In the separate $\Gamma = 5$ study, the main rule found no candidates after cycle-based background estimation. Two weaker candidates disappeared when the background or alignment changed. Cycle RMS declined by 4.29%, suggesting gradual evolution that warrants further study. Neither analysis confirmed a transient material event or its mechanism.

8.5 Level Selection, Validation and Main Conclusion

Level selection depends on the sampling rate and the target response. The nominal band of D_j is $[f_s/2^{j+1}, f_s/2^j]$. The index $s_j = 2^j \Delta t$ gives neither exact filter support nor event duration. Increasing f_s through interpolation does not increase measurement bandwidth.

Table 12: Selected components and their roles; no universal optimum is claimed.

Configuration	Reason
Synthetic STEP A_8	100 Hz: 2.56 s spacing for slow growth.
Real STEP A_{13}/A_5	Approximately 3,200 Hz: 2.56 s / 10 ms spacing for slow / early response; total depth 13.
Real LAOS D_6	48.89 Hz: nominal 0.382–0.764 Hz contains the third harmonic (0.477 Hz); total depth 8.

With 2,048 LAOS grid points, the third harmonic fell within nominal D7. D6 correlation decreased to 0.8814, while D7 was highly correlated at both grids. Given overlapping bands and boundary effects, correlation alone cannot establish D6 as the optimal level.

A3–A7 benchmark. A separate controlled test used known 20/150 ms rises, varied amplitudes/noise and paired dense/APR18-time sampling. Development used 120 positives/99 negatives; frozen testing used 240/200. The common interval was 0.275–0.995 s. Each rate threshold was calibrated on development negatives for a 5% false-alarm target; two consecutive exceedances were required. A hit meant the first candidate was within ±50 ms of onset. Selection used absolute error for hits and a fixed 0.2 s failure penalty.

Table 13: Frozen test under APR18 sampling. Timing errors describe hits only; misses remain essential to interpretation.

Feature	Hits / 240	False alarms / 200	Median hit error (ms)
A3 / A4	44 / 44	4 / 4	25.7 / 25.7
A5	42	4	26.0
A6	40	3	27.1
A7	62	5	25.8
Ordinary rate	44	4	25.7

The development comparison selected A7. Its penalised score was 154.4 ms, compared with 169.5 ms for A5, mainly because it had fewer failures rather than better timing among hits. A5 therefore remains a descriptive choice, not an established optimum. Even A7 matched only 25.8% of positives. The earliest real transient was outside the benchmark interval, and its onset error remains unknown.

Conclusion. DWT separates trends, periodic structure and fast activity, but does not automatically identify noise or material change. Reconstruction and sensitivity checks assess numerical consistency, not biological validity. Ordinary statistics often give similar information, and this study established no superiority over FFT/STFT/CWT. Further comparisons require matched smoothing, frozen detection rules and independent event references.

9 Conclusion

9.1 Main Findings

This project investigated change detection in blood rheology using conventional rheological analysis, physics-based modelling, and signal-processing techniques. Each approach provided a different perspective on changes in blood behaviour.

Conventional analysis examined changes in rheological properties through established mechanical indicators. Signal-processing methods, including FFT, STFT, CWT, and DWT, investigated frequency characteristics and localised variations in the measured signals. The t-ESSTV model provided a physics-based reference for stress prediction and residual analysis.

The t-ESSTV results showed strong agreement with APR18 steady-state measurements, while transient and LAOS predictions revealed limitations in reproducing time-dependent stress behaviour. In the semi-synthetic LAOS experiments, larger stress-amplitude perturbations were detected using rolling prediction errors, whereas the smallest perturbation did not trigger an alarm.

Together, these findings demonstrate the potential of complementary approaches for identifying rheological changes, while highlighting the challenges of reliable change detection in experimental blood measurements.

9.2 Comparison and Recommended Approach for Haemograph

The investigated methods provide different advantages for detecting changes in blood rheology. Conventional rheological analysis provides physically interpretable indicators, while FFT characterises the overall frequency and harmonic structure of measured signals. However, neither approach, as implemented in this project, provides the same direct time-frequency localisation as STFT and wavelet-based methods. The t-ESSTV model serves a separate role in interpreting stress responses and is therefore not included in this direct comparison of signal-based detection methods.

Among the signal-processing approaches, STFT and CWT demonstrated the clearest basis for direct comparison, as both were evaluated using the same controlled gain-change benchmark and comparable detection settings. CWT achieved earlier confirmation for some fundamental-amplitude and power features, whereas STFT performed better for several higher-harmonic detections. DWT provided useful multiscale representations of stress trends and periodic components, but its experiments did not establish a consistent detection advantage over simpler statistical measures.

Based on these findings, we recommend prioritising CWT as a candidate for further investigation by Haemograph, particularly for localised, multiscale analysis of rheological signals. However, the current evidence does not establish its overall superiority over STFT. Further evaluation under matched noise conditions, independent reference data, and consistent detection criteria is required before selecting a method for practical deployment.

9.3 Limitations

The main limitation of this study was the absence of independently verified coagulation onset times in the experimental datasets. Consequently, changes identified in measured signals could not be directly confirmed as coagulation events.

Although controlled synthetic and semi-synthetic experiments provided known change points for evaluation, these perturbations do not fully represent the complex physiological processes involved in blood coagulation. Furthermore, the findings were based on a limited range of experimental records and conditions, restricting conclusions about generalisation across donors and measurement protocols.

9.4 Future Work

Future work should prioritise collecting experimental measurements with independently verified change points to establish reliable ground truth. A broader evaluation across donors, shear protocols, and noise conditions would help assess the robustness and generalisability of the investigated methods.

Further work could also investigate more consistent detection thresholds and automated change-point identification. These developments would support the transition from exploratory analysis towards a more reliable change-detection framework for Haemograph's blood rheology measurements.

10 Data Science Contributions

10.1 Technical Contributions

This project involved the application of data science techniques to experimental blood rheology measurements. The team developed data-processing workflows, implemented conventional and signal-processing analyzes, and investigated the constitutive model t-ESSTV for stress prediction.

A further contribution was the construction and evaluation of controlled datasets with known change points, addressing the lack of ground truth in the original experimental measurements. The project also applied quantitative evaluation and visualisation techniques to examine prediction errors, signal changes, and detection performance.

These contributions established a foundation for evaluating change-detection methods in blood rheology and identified areas requiring further experimental validation.

10.2 Reproducibility and Project Handover

The project outputs include Python implementations of the analytical methods, data-processing and evaluation scripts, and visualisations of the experimental results. These computational workflows support reproducibility and provide a foundation for extending the analyses to additional rheological datasets.

The project code and related materials are available in the team's GitHub repository:

Group 35 GitHub Repository

The repository provides a central reference for the project's computational work and supports further evaluation and development by Haemograph.

References

[1] Paul S. Addison. "Wavelet transforms and the ECG: A review". In: *Physiological Measurement* 26.5 (2005), R155–R199. DOI: 10.1088/0967-3334/26/5/R01.

[2] Jont B. Allen and Lawrence R. Rabiner. "A unified approach to short-time Fourier analysis and synthesis". In: *Proceedings of the IEEE* 65.11 (1977), pp. 1558–1564. DOI: 10.1109/PROC.1977.10770.

[3] Matthew Armstrong and Andre Pincot. "Integration of thixotropy into Giesekus model for characterization of human blood". In: *AIP Advances* 11.3, 035029 (2021). DOI: 10.1063/5.0045578.

[4] Matthew Armstrong et al. "A simple approach for adding thixotropy to an elasto-visco-plastic rheological model to facilitate structural interrogation of human blood". In: *Journal of Non-Newtonian Fluid Mechanics* 290, 104503 (2021). DOI: 10.1016/j.jnnfm.2021.104503.

[5] Matthew Armstrong et al. "Recent advances in biofluid mechanics and bio- and hemorheology collating recent advances in predicting complex behavior of human blood with thixo-elasto-visco-plastic models and sequence of physical process". In: *Frontiers in Physics* 10, 889065 (2022). DOI: 10.3389/fphy.2022.889065.

[6] P. A. Evans et al. "Gel point and fractal microstructure of incipient blood clots are significant new markers of hemostasis for healthy and anticoagulated blood". In: *Blood* 116.17 (2010), pp. 3341–3346. DOI: 10.1182/blood-2010-02-269324.

[7] P. A. Evans et al. "Rheometrical detection of incipient blood clot formation by Fourier transform mechanical spectroscopy". In: *Journal of Non-Newtonian Fluid Mechanics* 148.1–3 (2008), pp. 122–126. DOI: 10.1016/j.jnnfm.2007.04.020.

[8] P. A. Evans et al. "Rheometry and associated techniques for blood coagulation studies". In: *Medical Engineering & Physics* 30.6 (2008), pp. 671–679. DOI: 10.1016/j.medengphy.2007.08.005.

[9] Frank A. Greco. "Reflectance spectroscopy of clotting blood: A description of the time-dependent behavior". In: *Archives of Pathology & Laboratory Medicine* 128.2 (2004), pp. 173–180. DOI: 10.5858/2004-128-173-RSOCBA.

[10] Rodrigo Capobianco Guido. "Wavelets behind the scenes: Practical aspects, insights, and perspectives". In: *Physics Reports* 985 (2022), pp. 1–23. DOI: 10.1016/j.physrep.2022.08.001.

[11] Fredric J. Harris. "On the use of windows for harmonic analysis with the discrete Fourier transform". In: *Proceedings of the IEEE* 66.1 (1978), pp. 51–83. DOI: 10.1109/PROC.1978.10837.

[12] Alexander Ya. Malkin and Svetlana R. Derkach. "Gelation of polymer solutions as a rheological phenomenon (mechanisms and kinetics)". In: *Current Opinion in Colloid & Interface Science* 73, 101844 (2024). DOI: 10.1016/j.cocis.2024.101844.

[13] Stephane G. Mallat. “A theory for multiresolution signal decomposition: The wavelet representation”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 11.7 (1989), pp. 674–693. DOI: 10.1109/34.192463.

[14] Andre Pincot and Matthew Armstrong. “Novel tensorial thixo-visco-plastic framework for rheological characterization of human blood”. In: *Scientific Reports* 11, 22004 (2021). DOI: 10.1038/s41598-021-01362-8.

[15] Andre Pincot et al. “Characterizing blood hysteresis via tensorial thixo-elasto-viscoplastic modeling”. In: *Physics of Fluids* 35.11, 113104 (2023). DOI: 10.1063/5.0169051.

[16] Peter J. Rousseeuw and Christophe Croux. “Alternatives to the median absolute deviation”. In: *Journal of the American Statistical Association* 88.424 (1993), pp. 1273–1283. DOI: 10.1080/01621459.1993.10476408.

[17] E. A. Ryan et al. “Structural origins of fibrin clot rheology”. In: *Biophysical Journal* 77.5 (1999), pp. 2813–2826. DOI: 10.1016/S0006-3495(99)77113-4.

[18] Vignesh V. Shanbhag et al. “Condition monitoring of hydraulic cylinder seals using acoustic emissions”. In: *The International Journal of Advanced Manufacturing Technology* 109 (2020), pp. 1727–1739. DOI: 10.1007/s00170-020-05738-4.

[19] A. Soleymani et al. “Characterizing change points and continuous transitions in movement behaviours using wavelet decomposition”. In: *Methods in Ecology and Evolution* 8.9 (2017), pp. 1113–1123. DOI: 10.1111/2041-210X.12755.

[20] Yang Yang et al. “Parameterised time-frequency analysis methods and their engineering applications: A review of recent advances”. In: *Mechanical Systems and Signal Processing* 119 (2019), pp. 182–221. DOI: 10.1016/j.ymssp.2018.07.039.

A Generative AI Usage Declaration

Generative artificial intelligence (GenAI) tools, including ChatGPT (OpenAI) and Codex, were used to support selected activities during the preparation of this project and report.

These tools were used to assist with understanding technical concepts, brainstorming methodological approaches, reviewing and debugging Python code, improving the clarity and grammar of written explanations, and refining the structure and presentation of the report.

All analytical methods, experimental results, interpretations, and conclusions were reviewed and evaluated by the project team. The team takes responsibility for the accuracy, integrity, and final content of the submitted report.